

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250284258

Kind Code

A1

Publication Date

September 11, 2025

Inventor(s)

Drilling; Jacob Charles et al.

DEVICE GROUP SYNCHRONIZATION

Abstract

Techniques for synchronizing device group data across different device control applications are described. A device group may include two or more smart home devices that may be controlled/supported by different device control applications. A user may set up device groups, including the same devices, in the different device control applications. A system may include a group synchronization service that synchronizes (e.g., merges) the multiple device groups across the different device control applications. The groups may be synchronized based on matching group names, matching devices included in the group, etc. After the device groups are synchronized, modifications made by the user to a device group in one device control application may be propagated to other device control applications.

Inventors: Drilling; Jacob Charles (Seattle, WA), Gupta; Amit (Seattle, WA), Patil; Rohit Ravindra (Seattle, WA), Aiken; Mark (Bainbridge Island, WA), Natraj; Vignesh Viswanat (Seattle, WA), Chorey; Michael (Bellingham, WA), Acharya; Parth Narendra (Austin, TX)

Applicant: Amazon Technologies, Inc. (Seattle, WA)

Family ID: 1000008629247

Appl. No.: 19/220229

Filed: May 28, 2025

Related U.S. Application Data

parent US continuation 17956004 20220929 parent-grant-document US 12353179 child US 19220229

us-provisional-application US 63390540 20220719

Publication Classification

Int. Cl.: G05B15/02 (20060101); G10L15/22 (20060101)

U.S. Cl.:

CPC G05B15/02 (20130101); G10L15/22 (20130101); G10L2015/223 (20130101)

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION [0001] This application is a continuation of, and claims priority to, U.S. non-provisional patent application Ser. No. 17/956,004, filed Sep. 29, 2022, and titled “GROUP DEVICE SYNCHRONIZATION,” which claims the benefit of and priority to U.S. Provisional Patent Application No. 63/390,540, filed Jul. 19, 2022, and entitled “SMART HOME MANAGEMENT WITH SPEECH CONTROL.” The above applications are herein incorporated by reference in their entireties.

BACKGROUND

[0002] Natural language processing systems have progressed to the point where humans can interact with computing devices using their voices and natural language textual input. Such systems employ techniques to identify the words spoken and written by a human user based on the various qualities of received input data. Speech recognition combined with natural language understanding processing techniques enable speech-based user control of computing devices to perform tasks based on the user's spoken inputs. Speech recognition and natural language understanding processing techniques may be referred to collectively or separately herein as spoken language understanding (SLU) processing. SLU processing may be used by computers, hand-held devices, telephone computer systems, kiosks, and a wide variety of other devices to improve human-computer interactions.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0003] For a more complete understanding of the present disclosure, reference is now made to the following description taken in conjunction with the accompanying drawings.

[0004] FIG. 1 is a conceptual diagram illustrating a system for synchronizing device group data across different device control systems, according to embodiments of the present disclosure.

[0005] FIG. 2 is a conceptual diagram of illustrating how different device groups may be merged into a single device group, according to embodiments of the present disclosure.

[0006] FIG. 3 shows example device group data that may be used by a system, according to embodiments of the present disclosure.

[0007] FIG. 4 is a signal flow diagram illustrating a process for modifying device group data based on a user input received at a device control application, according to embodiments of the present disclosure.

[0008] FIG. 5 is a signal flow diagram illustrating a process for modifying device group data based on a user input received at another device control application, according to embodiments of the present disclosure.

[0009] FIG. 6 is a conceptual diagram illustrating example system components that may be used to process a user input, according to embodiments of the present disclosure.

[0010] FIG. 7 is a conceptual diagram of components of a device, according to embodiments of the present disclosure.

[0011] FIG. 8 is a block diagram conceptually illustrating example components of a smart home

system, according to embodiments of the present disclosure.

[0012] FIG. 9 is a block diagram conceptually illustrating example components of a device, according to embodiments of the present disclosure.

[0013] FIG. 10 is a block diagram conceptually illustrating example components of a system, according to embodiments of the present disclosure.

[0014] FIG. 11 illustrates an example of a computer network for use with the overall system, according to embodiments of the present disclosure.

DETAILED DESCRIPTION

[0015] Automatic speech recognition (ASR) is a field of computer science, artificial intelligence, and linguistics concerned with transforming audio data associated with speech into a token or other textual representation of that speech. Similarly, natural language understanding (NLU) is a field of computer science, artificial intelligence, and linguistics concerned with enabling computers to derive meaning from natural language inputs (such as spoken inputs). ASR and NLU are often used together as part of a language processing component of a system. Text-to-speech (TTS) is a field of computer science concerning transforming textual and/or other data into audio data that is synthesized to resemble human speech. Natural language generation (NLG) is a field of artificial intelligence concerned with automatically transforming data into natural language (e.g., English) content.

[0016] An environment (e.g., a house, an apartment, an office, a store, a hotel lobby, etc.) may include one or more smart home devices. As used herein, a “smart home device” refers to a device that may be controlled by another device or system in response to receiving a user input (e.g., a spoken input, an input via a GUI, a gesture input, etc.). Example smart home devices include, but are not limited to, light switches, TVs, plugs, outlets, light bulbs, motion sensors, speakers, door locks, window locks, garage doors, security cameras, home monitoring devices, ovens, microwaves, washers, dryers, temperature sensors, and thermostats.

[0017] An environment can include smart home devices that are controlled by different systems. Some smart home devices may be controlled using a first device control system, via a first device control application provided by the first system, while other smart home devices may be controlled using a second, different, device control system, via a second device control application provided by the second system. For example, a user's house may include an outlet that can be controlled using an Amazon Alexa application and light bulbs that can be controlled using a Philips Hue application. In this example, Amazon Alexa may represent one device control system and Philips Hue may represent another device control system.

[0018] A user can group two or more smart home devices, which is referred to herein as a “device group”, to control the devices in the group together (e.g., using one input, operating in a synchronized manner, etc.). The devices may be of different types. For example, a first device group may include light bulbs and an outlet, and the user may provide a single input that causes the light bulbs and the outlet in the first device group to turn on. As another example, a second device group may include a right speaker, a left speaker, and a sub-woofer, and the second device group may output audio in a synchronized manner.

[0019] A device group can include smart home devices that are controlled by different systems. When the device group includes devices controllable by different systems, users may have to manage the device group in the corresponding different applications installed at a user device(s). For example, if a user created a device group using a first device control application and a second device control application, any time the user modifies the device group (e.g., changes the group name, adds/removes devices, changes device settings, changes group settings, etc.), the user has to separately cause data to be input into the first and second device control applications in order to carry through the group change to both applications. This can result in a fragmented experience, in particular, when device group information (e.g., group name, devices within the group, etc.) is different in different applications.

[0020] The present disclosure relates to techniques for synchronizing device group data across multiple different device control systems. In some embodiments, after a user approves device group synchronization across different device control systems, a system requests the device control systems to provide device group data associated with the user, and the system processes various device group data, associated with different device control systems, to determine whether two or more device groups correspond to one another and should be merged. The techniques described herein enable synchronization of device group names and device group members based on, for example, determining that the device group names are the same or similar, and/or that the device group members are the same or similar (for example based on device identifiers or the like). For example, a device group named “Living Room” in a first device control system and a device group named “livingroom” in a second device control system may be determined to correspond to each other. As another example, a first device group including devices A, B, C may be determined to correspond to a second device group including the same devices A, B, C.

[0021] Device groups from different device control systems may be merged to form one device group, or may be kept as separate device groups. In some cases, two device groups that are merged may have the same number of members that correlate with the same devices. In other cases, two device groups that are merged may have different number of members or correlate with different devices, and the merged group may include devices from both groups.

[0022] The system described herein can also propagate modifications to a device group that are made by a user in one device control system to other device control systems that the user uses to control smart home devices, without the user expressly replicating the changes in the other device control applications.

[0023] Teachings of the present disclosure provide, among other things, an improved user experience by synchronizing device group information across different device control systems, and enabling control of smart home devices, supported by different device control systems, via one of the device control application.

[0024] A system according to the present disclosure may be configured to incorporate user permissions and may only perform activities disclosed herein if approved by a user. As such, the systems, devices, components, and techniques described herein would be typically configured to restrict processing where appropriate and only process user data in a manner that ensures compliance with all appropriate laws, regulations, standards, and the like. The system and techniques can be implemented on a geographic basis to ensure compliance with laws in various jurisdictions and entities in which the components of the system and/or user are located.

[0025] FIG. 1 illustrates a system **100** for synchronizing device group data across different device control applications/systems, according to embodiments of the present disclosure. The system **100** may include a user device **110**, local to a user **105**, in communication with a smart home supporting device(s) **130** via a network(s) **199**. The system **100** may further include one or more smart home devices **112**, such as a light bulb **112a**, an outlet **112b**, a light switch **112c**, etc. One or more of the smart home device(s) **112** may be in communication with the smart home supporting device(s) **130**, the user device **110**, and/or the device control supporting device(s) **135** via the network(s) **199**. The network(s) **199** may include any combination of the Internet and/or any other wide- or local-area network, and may include wired, wireless, and/or cellular network hardware. In some embodiments, one or more of the smart home devices **112** may be in communication with the user device **110** via a private network (e.g., a home network), Bluetooth, Near Field Communication (NFC), etc. Although the figures and discussion of the present disclosure illustrate certain steps in a particular order, the steps described may be performed in a different order (as well as certain steps removed or added) without departing from the present disclosure.

[0026] The user device **110** may include or may be in communication with one or more components that enable the user **105** to operate the smart home device(s) **112**. Some smart home devices **112** may also include components that enable the user **105** to operate the smart home

device **112**. A device group, as used herein, may include a combination of user devices **110** and/or smart home devices **112**. Example smart home devices **112** include, but are not limited to, light switches, TVs, plugs, outlets, light bulbs, motion sensors, speakers, door locks, window locks, garage doors, security cameras, home monitoring devices, ovens, microwaves, washers, dryers, temperature sensors, and thermostats. The smart home devices **112** may also include smart devices outside a home that may be in a different environment (such as an office, a hotel room, a lobby, a store, a vehicle, etc.). Control of the smart home devices **112** may include control of device components attached to or in communication with the smart home device **112**, such as, device sensors (e.g., gas tank sensor, tire pressure sensor, fridge door sensor, front door sensor, temperature sensors, etc.), device timers (e.g., washing machine timer, oven timer, etc.), and other components (e.g., washing machine settings, vehicle speakers, vehicle controls, etc.).

[0027] More than one environment, which may be referred to as “premises”, may be associated with the user **105**, where individual premises may include different sets of smart home devices **112** and/or user devices **110**. For example, the user **105** may have a primary home, including a first set of devices, a vacation home including a second set of devices, a vehicle including a third set of devices, etc.

[0028] The smart home supporting device(s) **130** may include various components. With reference to FIG. 1, the smart home supporting device(s) **130** may include a device control application **140**, a device group storage **165**, and a group synchronization service (GSS) **145**. The device control application **140** may receive user inputs, via, for example, a corresponding application on the user device **110**, that enables control of the device(s) **110**, **112**. The corresponding application on the user device **110** may provide a user interface for the user **105** to provide various inputs, such as authorization for synchronizing device groups, adding or removing devices **110**, **112**, modifying settings for the devices **110**, **112**, etc. In some embodiments, some of the functionalities described as being performed by the device control application **140** may be performed by the corresponding application on the user device **110**. The device control application **140** may be a first device control application that can be used to control a particular set of devices **110**, **112**. Control of a device **110**, **112** by the device control application **140** may depend on device configurations for the device **110**, **112**, such as device type, device manufacturer, device settings, etc.

[0029] The system **100** may also include one or more device control supporting device(s) **135**, which are in communication with the smart home supporting device(s) **130**. The device control supporting device(s) **135** may also be in communication with the user device **110** via the network(s) **199** (although not shown). The device control supporting device(s) **135** may include or support another device control application(s) (e.g., device control application **450** shown in FIG. 4), separate from the device control application **140**, that can be used to control another set of devices **110**, **112** (depending on, for example, device configurations for the device **110**, **112**). For example, a first smart home device **112a** may be controllable using the device control application **140**, a second smart home device **112b** may be controllable using the device control application **450** of the device control supporting device(s) **135**, etc. Some devices may be controllable using more than one device control application, depending on how the user **105** configures the device. The functionalities of the device control supporting device(s) **135** may be provided as a skill component (e.g., skill component(s) **690**, skill supporting device(s) **625**), an application, or other system components,

[0030] The user device **110** may include a first application corresponding to the smart home device supporting device(s) **130** that enables the user **105** to control first smart home devices **112** (e.g., Amazon Alexa compatible devices). The user device **110** may further include a second separate application corresponding to the device control supporting device(s) **135** that enables the user **105** to control second smart home devices **112** (e.g., Philips Hue compatible devices). In some embodiments, the first smart home devices **112** and the second smart home devices **112** may include overlapping devices, for example, a light bulb **112a** that can be controlled using both of the

supporting device(s) **130** and **135**. Some operations of the smart home devices **112** may be enabled via the application(s) on the user device **110**, without communications needed with the supporting devices **130**, **135**. For example, the user **105** may select a GUI button presented via a user interface of the application at the user device **110** to turn on the light bulb **112a**. In some embodiments, the applications on the user device **110** may also perform functions of synchronizing device groups, as described herein.

[0031] Referring to FIG. **1**, the user **105** may provide an input using the user device **110**, where such input may be a spoken input, a typed input, selection of GUI element, a gesture input, etc. In some embodiments, the user device **110** may be displaying a user interface (UI) screen corresponding to the device control application **140**, where the UI screen may be requesting permission, from the user **105**, to synchronize device groups across different device control applications.

[0032] In some embodiments, the UI screen may show data identifying the different device control applications available in the system **100** for device group synchronization, where such device control applications may be systems, skills or other components (e.g., the device control supporting device(s) **135**) that are registered or otherwise configured to operate in conjunction with the smart home supporting device(s) **130**. In other embodiments, the UI screen may show data identifying the device control applications that the user **105** has access to based on, for example, the user **105** setting up/connecting the device control applications to a user profile of the user **105**. The user profile may be stored in a profile storage **670/770** described below. In yet other embodiments, the device control applications shown to the user **105** may be ones that correspond to the smart home devices **112** within the user's environment.

[0033] The user device **110** may send (step **1** of FIG. **1**) input data to the device control application **140**, where the input data represents permission from the user **105** to synchronize device group data across different device control applications. The device control application **140** may send (step **2**) the input data to a group discovery component **150** of the GSS component **145**. In response to receiving the input data representing the permission, the GSS component **145** may initiate a device group synchronization process.

[0034] The device group synchronization process may involve the group discovery component **150** to send requests to one or more of the device control supporting device(s) **135**. For example, the group discovery component **150** may send (step **3a**) a first request to the device control supporting device(s) **135a** for a first device control application, and may send (step **3b**) a second request to the device control supporting device(s) **135b** for a second device control application. The first request may be a request for device group data corresponding to one or more device groups associated with the user **105**. The first request may include a user profile identifier corresponding to the user **105**, where the user profile identifier may be used by the device control supporting device(s) **135a** to identify the user **105**. In some cases, the user profile identifier in the first request may be different than a user profile identifier used by the smart home supporting device(s) **130** (and a supporting device(s) **120** shown in FIGS. **6** and **7** and described below) to identify the user **105**. The smart home supporting device(s) **130** may store data representing user profile identifiers for the user **105** for the different device control supporting device(s) **135**. Similarly, the second request may include a user profile identifier corresponding to the user **105** and the device control supporting device(s) **135b**. In some embodiments, the group discovery component **150** may send API requests to the device control supporting device(s) **135**. The group discovery component **150** may send the requests to the device control supporting device(s) **135** associated with the user **105**.

[0035] In response to receiving the first request, the device control supporting device(s) **135a** may send (step **4a**) first device group data corresponding to at least a first device group, where the first device group data may be received by an inbound events component **155** of the GSS **145**. The first device group data may include a first device group identifier used by the device control supporting device(s) **135a** to identify the first device group. The first device group data may also include one

or more device identifiers (a first set of device identifiers) associated with (a first set of) smart home devices included in the first device group. The first device group data may also include a first group name, which may be a user-provided name for the first device group.

[0036] In response to receiving the second request, the device control supporting device(s) **135b** may send (step **4b**) second device group data corresponding to at least a second device group, and the second device group data may be received by the inbound events component **155**. The second device group data may include a second device group identifier used by the device control supporting device(s) **135b** to identify the second device group. The second device group data may also include one or more device identifiers (a second set of device identifiers) associated with (a second set of) smart home devices included in the second device group. The second device group data may also include a second group name, which may be a user-provided name for the second device group.

[0037] The inbound events component **155** may send (step **5**) the first device group data and the second device group data (and other device group data received from other device control supporting device(s) **135** in response to requests from the group discovery component **150**) to a group merge component **152** of the GSS **145**. The group merge component **152** may be configured to determine device groups, from different device control applications, that correspond to one another, and merge them so that the device groups are synchronized across the different device control applications.

[0038] For merging the device groups, the group merge component **152** also retrieves (step **6**) (third) device group data from the device group storage **165**. The device group storage **165** may store device group data based on device groups created by the user **105** using the device control application **140**/the smart home supporting device(s) **130**. The third device group data may include a third device group identifier used by the smart home supporting device(s) **130** to identify the third device group. The third device group data may also include one or more device identifiers (a third set of device identifiers) associated with (a third set of) smart home devices included in the third device group. The third device group data may also include a third group name, which may be a user-provided name for the third device group.

[0039] The techniques described herein allow a user to synchronize device groups across multiple applications (e.g., Amazon Alexa, Philips Hue, Samsung Smart Things, etc.) without the user having to make the same change on all the applications. There may be conflicting or different device group information between the various device control applications, or device groups may share/have the same names but represent different groups in a user's environment. This can cause friction (e.g., undesired experience) for users as they may not be able to target the right/desired group in a particular application. The group merge component **152** may implement logic to handle device group conflicts to merge the device groups.

[0040] The group merge component **152** may process the (e.g., the first, the second and the third) device group data provided by one or more device control supporting device(s) **135** and the device group storage **165**. The group merge component **152** may use one or more rules-based engines, one or more machine learning models, one or more statistical models, and/or other mechanisms to process the device group data and identify correlated groups (i.e., groups from different device control applications that are intended to control similar groupings of devices and/or spaces).

[0041] In some embodiments, the group merge component **152** may determine whether two (or more) device groups correspond to one another based on processing the respective device group names (provided by the user **105**) for the device groups. In an example embodiment, the group merge component **152** may use a name matching algorithm, for example, the Levenshtein Distance algorithm (also known as the Edit distance-based algorithm). In example embodiments, the Levenshtein Distance algorithm may compute the number of edits required to transform one string (e.g., a first device group name) to another (e.g., a second device group name). The edits count the following as an operation: insertion of a character; deletion of a character; and substitution of a

character. The larger the number of operations to match the strings, the less the similarity between the two strings. The Levenshtein Distance algorithm can be used to measure the similarity between the device group names of different lengths well. Another (or additional) example algorithm that may be implemented by the group merge component **152** may be the Jaccard Similarity algorithm, which involves determining a coefficient that measures the similarity between finite sample sets, and is defined as the size of the intersection divided by the size of the union of the sample sets. The sample sets may correspond to common device group names (e.g., living room, kitchen, hotel lobby, dining room, bathroom, children's room, [person's name] room, outdoors, hallway, etc.). Another (or additional) example algorithm that may be implemented by the group merge component **152** may be the Normalized Hamming Distance algorithm, which involves determining a percentage representing a dissimilarity of two strings/device group names. A lower value of Normalized Hamming distance means the two strings are more similar.

[0042] In some embodiments, the group merge component **152** may determine that two device group names correspond to one another based on a similarity (e.g., character similarity, semantic similarity, or the like) of the device group names. For example, the device group names “Living Room” and “Family Room” may be determined to correspond to the same device group.

[0043] The group merge component **152** may determine whether smart home devices in a device group correspond (e.g., match, are the same, are similar, etc.) to the smart home devices in another device group. Such determination may be based on a device type of the smart home device, a user-provided name for the smart home device, a device identifier for the smart home device, etc. In analyzing the device names, the group merge component **152** may use a name matching algorithm, a semantic similarity algorithm, a rules-based engine, and/or other models described herein. As an example, a first device name “bedroom lights” and a second device name “bedroom light” may be determined to correspond to the same smart home device.

[0044] These merge techniques can be used in varying combinations, priorities, and hierarchies. For example, where groups from at least one device control application have generic names like “Group 1” and “Group 2” these names may be discounted for purposes of matching and the names of the devices can be matched against each other or the groups of another device control application. For example, if the devices in “Group 1” include “bedroom light” then the name of the device can be matched against a “bedroom” group from the other device control application. Similar matching can be done across group and device sets such that similarities that meet a defined threshold can be determined to be ‘matches.’

[0045] The group merge component **152** may implement rules to merge two device groups. Some example rules and scenarios are described below using “G” to denote a device group and (A, B, C, D, X, etc.) to denote smart home devices included in a group G.

[0046] In one example scenario, the device group storage **165** may not have a device group corresponding to a first device group provided by the device control supporting device(s) **135a**. For example, the device control supporting device(s) **135a** may send data indicating group G1=(A, B, C), and the merged group may be G1=(A, B, C). The GSS **145**, in this scenario, may implement a rule that a merged group is based on a received group information where there is no prior matching group in the device group storage **165**. The GSS **145** may also create a device group in the smart home supporting device(s) **130** by storing, in the device group storage **165**, device group data representing G1=(A, B, C). The device group data may include a device group identifier, a set of device identifiers corresponding to the smart home devices A, B, C, a user-provided device group name, and/or user-provided device names. The device group data may also include a device group identifier used by the device control supporting device(s) **135a** to identify the group G1.

[0047] In another example scenario, the device control supporting device(s) **135a** may provide a device group including a device (device C) that is not included in a device group provided by the device group storage **165**, but the device C is identifiable by the smart home supporting device(s) **130**. For example, the device group storage **165** may provide G1=(A, B) and the device control

supporting device(s) **135a** may provide $G2=(A, B, C)$. The group merge component **152** may determine the merged group $G3=(A, B, C)$ based on the device C being identifiable (e.g., the device being connected to the user profile of the user **105**) by the smart home supporting device(s) **130**. The GSS **145**, in this scenario, may implement a rule that a merged group includes a device in received group information where there is no matching device in a group in the device group storage **165**, and the device is discoverable by the smart home supporting device(s) **130**. The GSS **145** may store, in the device group storage **165**, device group data representing the group $G3$.

[0048] In yet another example scenario, the device control supporting device(s) **135a** may provide a device group including a device (device D) that is not included in a device group provided by the device group storage **165**, and the device D is not identifiable by the smart home supporting device(s) **130**. For example, the device group storage **165** may provide $G1=(A, B)$ and the device control supporting device(s) **135a** may provide $G2=(A, B, D)$. The group merge component **152** may determine the merged group $G3=(A, B)$ based on the device D not being identifiable by the smart home supporting device(s) **130**. In this scenario, the GSS **145** may implement a rule that a merged group does not include a device in received group information where there is no matching device in a group in the device group storage **165** and the device is not discoverable by the smart home supporting device(s) **130**. Other components of the smart home supporting device(s) **130** may perform certain operations to identify the device D (e.g., request the user **105** to add the device D to the user profile, request the user to confirm device D is to be added, request the user **105** to add an application/skill that controls the device D, etc.). The GSS **145** may not store additional data in the device group storage **165**, since the group $G1$ already matched the merged group $G3$. The GSS **145** may communicate device group data representing the group $G3$ to the device control supporting device(s) **135a** as described below via an outbound events component **160**.

[0049] In yet another example scenario, the device group provided by the device group storage **165** may include an additional device than the device group provided by the device control supporting device(s) **135a**. For example, the device group storage **165** may provide $G1=(A, B, C)$ and the device control supporting device(s) **135a** may provide $G2=(A, B)$. The group merge component **152** may determine the merged group $G3=(A, B, C)$. In this scenario, the GSS **145** may implement a rule that a merged group includes a device in a group in the device group storage **165** where there is no matching device in a received group information. The GSS **145** may not store additional data in the device group storage **165**, since the group $G1$ already matches the merged group $G3$. The GSS **145** may communicate device group data representing the group $G3$ to the device control supporting device(s) **135a** as described below.

[0050] In another example scenario, the device groups provided by the device group storage **165** and the device control supporting device(s) **135a** may have the same group name, but may include different devices. For example, the device group storage **165** may provide $G1=(X)$ and the device control supporting device(s) **135a** may provide $G2=(A, B)$. The group merge component **152** may determine the merged group $G3=(A, B, X)$. As another example, the device group storage **165** may provide $G1=(A, B, C)$ and the device control supporting device(s) **135a** may provide $G2=(C, D, E)$. The group merge component **152** may determine the merged group $G3=(A, B, C, D, E)$. The GSS **145**, in these scenarios, may implement a rule that a merged group includes devices in received group information and devices in a group in the device group storage **165**, when the group names are the same (e.g., are an exact word match, are a semantic match, match to a certain extent (e.g., 98% matching), etc.) The GSS **145** may store, in the device group storage **165**, device group data representing the group $G3$, and may communicate the device group data representing $G3$ to the device control supporting device(s) **135a**.

[0051] In another example scenario, the device groups provided by the device group storage **165** and the device control supporting device(s) **135a** may have different device group names, but may include the same devices. For example, the device group storage **165** may provide $G1=(A, B, C)$ and the device control supporting device(s) **135a** may provide $G2=(A, B, C)$. The group merge

component **152** may determine the merged group $G3=(A, B, C)$. In some cases, the merged group **G3** may retain the **G1** group name. In other cases, the device control application **140** may ask the user **105** to provide a name for the merged group **G3**. The GSS **145**, in this scenario, may implement a rule that groups are merged when they include the same devices. The GSS **145** may store, in the device group storage **165**, device group data representing the group **G3** with the group name, and may communicate the device group data representing **G3** and the group name to the device control supporting device(s) **135a**.

[0052] In yet another example scenario, the device group storage **165** may include a first device group with a first name and a second device group with a second name, and the device control supporting device(s) **135a** may provide a device group with the same/similar name as the second device group, but including devices of the first device group. For example, the device group storage **165** may provide $G1=(A, B)$ and $G2=(C, D)$, and the device control supporting device(s) **135a** may provide $G3=(A, B)$ that has the same name as the group **G2**. The group merge component **152** may determine the merged group as $G4=(A, B)$, which may be communicated to the device control supporting device(s) **135a** as replacing the group **G3**, and may be used to replace **G1** in the device group storage **165**. In this scenario, the GSS **145** may implement a rule that groups are merged based on included devices, rather than group names.

[0053] The group merge component **152** may also determine the name for the merged group. In one example, the group merge component **152** may use the group name provided by the device group storage **165**. In another example, one of the device group names is misspelled, then the properly spelled device group name may be used for the merged group (e.g., “livingroom” vs. “living room”). Any set of rules to prefer one device group name based on source, format, specificity, or similar aspects can be defined and applied by the GSS **145**.

[0054] In some cases, when the device group names are not the same/similar, and/or the devices in the groups are not the same/similar, then the GSS **145** may ask the user **105** to confirm whether two device groups are to be merged.

[0055] In some embodiments, the group merge component **152** may also select/reconcile device names for the devices in the merged group, when the devices have different names in group data received from the device control supporting device(s) **135** and the device group storage **165**. In some embodiments, the group merge component **152** may implement a rule that the merged group includes the device name that is included in the group of the device group storage **165**. In other embodiments, the group merge component **152** may implement a rule that selects the device name that does not have spelling mistakes. In yet other embodiments, when two (or more groups) do not include the same devices, the group merge component **152** may select the device name that is available. In some cases, the GSS **145** may ask the user **105** to confirm or provide the device name.

[0056] The group merge component **152** may send (step **7**), to the device group storage **165**, (merged) device group data representing the merged device group. The merged device group data may include a device group identifier, a set of device identifiers corresponding to the set of smart home devices included in the merged device group, and a device group name. In some cases, the device group identifier may be the same device group identifier as the one provided by the device group storage **165** (in step **6**), and the device group data corresponding to the device group identifier may be updated based on the merged device group data. For example, the smart home devices associated with the device group identifier may be updated based on a device(s) being added or removed from the original device group.

[0057] The group merge component **152** may send (step **8**) the merged device group data to the outbound events component **160** for sending to the device control supporting device(s) **135** to synchronize device group data between the smart home supporting device(s) **130** and the device control supporting device(s) **135**. The outbound events component **160** may send (step **9a**), to the device control supporting device **135a**, the merged device group data, and may send (step **9b**), to the device control supporting device **135b**, the merged device group data. The device control

supporting device(s) **135** may store the merged device group data, and may associate a device group identifier with the merged device group data. In some cases, the device control supporting device(s) **135** may replace existing device group data, for example, the device group data communicated in step **4**, or may create a new device group and delete the original device group. The outbound events component **160** may send the merged device group data using an API request. [0058] In some embodiments, the device control supporting device(s) **135** may communicate, to the GSS **145**, a device group identifier used to identify the merged device group. The GSS **145** may store the received device group identifier in the device group storage **165** as an identifier used by the device control supporting device(s) **135**.

[0059] In communicating data, such as device group data and/or device identifiers, the smart home supporting device(s) **130** may use a “directed ID” that may be specific to and that may be used by the particular device control supporting device(s) **135** to identify the device group and/or device. For example, when sending the merged device group data to the device control supporting device(s) **135a**, the message may include a first directed ID that the device control supporting device(s) **135a** may use to identify the device group; and when sending the merged device group data to the device control supporting device(s) **135b**, the message may include a second directed ID that the device control supporting device(s) **135b** may use to identify the device group. In some embodiments, portions of the merged device group data may be communicated to the device control supporting device(s) **135**, where the portions may correspond to a device(s) associated with/in control of the particular device control supporting device(s) **135**.

[0060] In some embodiments, the device control supporting device(s) **135** may be implemented as a skill supporting device(s) **625** described below (e.g., the device control supporting device(s) **135a** may be a first skill supporting device(s) **625a**, the device control supporting device(s) **135b** may be a second skill supporting device(s) **625b**, etc.). In some embodiments, the smart home supporting device(s) **130** may communicate with a skill component **690/790**, which may in turn communicate with the skill supporting device(s) **625**/the device control supporting device(s) **135**. The user **105** may provide permissions enabling the individual skill components to communicate with the smart home supporting device(s) **130** to control smart home devices associated with the particular skills, to enable the skills to respond to user inputs provided by the user **105** to the smart home supporting device(s) **130**, and to perform other functionalities in conjunction with the smart home supporting device(s) **130** and/or the supporting device(s) **120**.

[0061] In some embodiments, the device control supporting device(s) **135**/the skill component **690** may request permission from the smart home supporting device(s) **130** and/or the user **105** (via a corresponding device control application) to synchronize device group data. In some embodiments, the user **105** may be notified, via the device control application **140** and the user device **110**, when the device control supporting device(s) **135** requests such permissions. In some embodiments, the user **105** may provide permissions (without prompting from a system) to synchronize device group data between different device control applications. In some embodiments, the user **105** can disable synchronization of device group data without having to disable other functionalities of the device control supporting device(s) **135**/the skill.

[0062] In some embodiments, the device control supporting device(s) **135**/skill may use (at least) two types of permissions—one for synchronizing device groups and one for synchronizing premises. As used herein, premises refers to different environments associated with the user **105** via the same user account/profile, for example, a primary home, a vacation home, an office, etc. The device group synchronization permissions allow the device control supporting device(s) **135** to retrieve device groups that contain devices reported/supported by the device control supporting device(s) **135**. The premises synchronization permissions allow the device control supporting device(s) **135** to retrieve information for all premises on a user account of the user **105**, and allow retrieval of devices reported by their skill. Such permissions allow a skill developer to request access to a resource and can optionally require user approval to access the resource. The permissions also

allow the user to toggle (turn on and turn off) the permissions without disabling the entire skill. For example, the user **105** could disable the synchronization permission for a particular device control skill, but still use that device control skill's smart home devices.

[0063] In some embodiments, the GSS **145** may be configured to synchronize premises group information between various device control applications. The premises group information may include information related to devices included in the premises, such as, premises name, device names, device identifiers, device location (e.g., within the premises), etc. Different device control applications may include different (or the same) premises group information, and the GSS **145** may synchronize the information in a similar manner as described herein with respect to synchronizing device group information. In synchronizing premises group information, the GSS **145** may merge data relating to premises name, device names, device locations, etc. For example, the device control supporting device(s) **135** may send first premises information associated with a first premises name “vacation” and including a first device “living room TV” located in “living room,” and a second device “smart speaker” located in “kitchen.” Continuing with the example, the device group storage **165** may include second premises information associated with a second premises name “vacation home” and including a first device “TV for living room” located in “living room,” and a second device “speaker” located in “kitchen.” The GSS **145**, in this example, may determine merged premises group information associated with a premises name “vacation home” and including a first device “living room TV” located in “living room” and a second device “speaker” located in “kitchen.” Such merged premises group information may be communicated to the device control supporting device(s) **135** and/or may be stored/updated in the device group storage **165**. Synchronizing the premises group information may enable the user **105** to update information related to a device **110/112** in one device control application, and have the system automatically synchronize it to other device control application(s). For example, the user **105** may move a device named “speaker” from location “kitchen” to location “dining room.” The change in location may be manually entered by the user **105** via the device control application **140** (or the device control application **450**) or may be automatically recognized by the smart home supporting device(s) **130** (or the device control supporting device(s) **135**). The change in location may be communicated to the other device control application as described herein. In some embodiments, the inbound events component **155** and the outbound events component **160** may be implemented at the smart home supporting device(s) **130** separately from the GSS **145**.

[0064] FIG. 2 is a conceptual diagram of illustrating how different device groups may be merged into a single device group, according to embodiments of the present disclosure. A device group can include two or more devices (which may sometimes be referred to as “members”), such as user devices **110** and/or smart home devices **112**. The devices in a device group could be controlled by different device control applications. For example, the speech controllable device **110a** and the plug **112c** shown in FIG. 2 may be controlled by the device control application **140**/the smart home supporting device(s) **130**, while the light bulbs **112a**, **112b** may be controlled by the device control supporting device(s) **135a**. FIG. 2 shows a first device group **210** including devices **110a**, **112a-112c**, and the device group storage **165** may store first device group data corresponding to the first device group **210**. The first device group data may include a first group identifier for the first device group **210**, and may include a first device identifier associated with the device **110a**, a second device identifier associated with the device **112a**, a third device identifier associated with the device **112b**, and a fourth device identifier associated with the device **112c**. The first device group data may also include a user-provided group name (e.g., “Living Room”) for the first device group **210** and user-provided device names (e.g., “Speaker”, “Lights”, “Back Lights”, “Fan”) for the individual devices **110a**, **112a-112c**.

[0065] FIG. 2 shows a second device group **220** including devices **110a**, **112a**, and **112c**, and the device control supporting device(s) **135** may send second device group data corresponding to the second device group **220**. The second device group data may include a second group identifier for

the second device group **220**, and may include a fifth device identifier associated with the device **110a**, a sixth device identifier associated with the device **112a**, and a seventh device identifier associated with the device **112c**. The second device group data may also include a user-provided group name (e.g., “livingroom”) for the second device group **220** and user-provided device names (e.g., “Speakers”, “Light bulb”, “Fan”) for the individual devices **110a**, **112a**, and **112c**. [0066] As shown in FIG. 2, a merged device group **230**, based on merging the first device group **210** and the second device group **220**, may include the devices **110a**, **112a-112c**. Such merging may be performed based on the user-provided group names, such as “Living Room” and “livingroom”, being similar or the same. As shown, the merged device group **230** may be named “Living Room” based on a rule that the correct spelling of a word/phrase is used for the merged group and/or a rule that the group name included in the device group storage **165** takes precedent over group names from other device control applications.

[0067] The merging of the two device groups shown in FIG. 2 may also be based on the user-provided device names being similar or the same. As shown and described herein, two device groups may be merged even though the groups have different number of devices or different types of devices. In the example shown in FIG. 2, the device **110a** is the same in the two device groups **210**, **220**; the system may make that determination based on the user-provided device name “Speaker” and/or a mapping between the first device identifier and the fifth device identifier. The device **112a** may be determined to be the same in the two device groups **210**, **220** based on the device names “Lights” and “Light bulb” being similar, and the device **112c** may be determined to be the same based on the device name “Fan” being the same in the two device groups **210**, **220**. As shown, the device names in the merged device group **230** may be “Speaker”, “Lights”, “Back Lights” and “Fan”, which may be based on a rule that the device name included in the device group storage **165** takes precedent over device names from other device control applications.

[0068] FIG. 3 shows example device group data that may be processed by the GSS **145** to determine merged group data. First device group data **310** may be provided by the device group storage **165**, and second device group data **320** may be provided by the device control supporting device(s) **135a**. The group merge component **152** may determine merged device group data **330** based on the first device group data **310** and the second device group data **320**. The first and second device groups (corresponding to **310** and **320**) may be merged based on the group name “Living Room” matching. As shown, the merged device group data **330** may include the group name (**334**) “Living Room.” The merged device group data **330** may be stored in the device group storage **165**, and may include the same group id (**332**) as the first device group data **310**. As shown, the merged device group data **330** may also include a group id for the other device control application (**336**), which is the same group id included in the second device group data **320**. The merged device group data **330** also includes device ids and device names from the first device group data **310**, and also includes device ids **338** from the second device group data **320** corresponding to the other device control application [App Name]. Using at least the data **336** and **338**, the smart home supporting device(s) **130** can enable control of smart home devices associated with and supported by the device control supporting device(s) **135a**. When a user input is received, the smart home supporting device(s) **130** may communicate such user input to the device control supporting device(s) **135a** along with the group id **336** (and optionally the device ids **338**), so that the device control supporting device(s) **135a** can identify within its system which device group the user input relates to. The merged device group data **330** may represent a relationship, a correlation or otherwise an association between the two device groups, and as such, the merged group data **330** may include a first record including data (e.g., group id, user provided group name, device ids, user provided device names, etc.) corresponding to the smart home supporting device(s) **130** and a second record including data (e.g., group id, user provided group name, device ids, user provided device names, etc.) corresponding to the device control supporting device(s) **135a**.

[0069] Merged device group data, such as data **330**, may include merged group data correlating

device groups and devices across any number of different device control applications. In some embodiments, the merged device group data may also include the user-provided group names and user-provided device names, which may differ for individual device control applications. [0070] FIG. 4 is a signal flow diagram illustrating a process for modifying device group data based on a user input received at a device control application **450** of the device control supporting device(s) **135**, according to embodiments of the present disclosure. An application corresponding to the device control application **450a** may be installed on the user device **110**, and may provide a user interface for the user **105** to provide various inputs, such as authorization for synchronizing device groups, adding or removing devices, modifying settings for devices, etc. In some embodiments, some of the functionalities described as being performed by the device control application **450** may be performed by the corresponding application on the user device **110**. The user device **110** may include a first application corresponding to the smart home device supporting device(s) **130**/the device control application **140** that enables the user **105** to control first devices **110**, **112** (e.g., Amazon Alexa compatible devices). The user device **110** may further include a second separate application corresponding to the device control supporting device(s) **135**/the device control application **450** that enables the user **105** to control second devices **110**, **112** (e.g., Philips Hue compatible devices).

[0071] The device control application **450a** may be implemented at the device control supporting device(s) **135a**, and a device control application **450b** may be implemented at the device control supporting device(s) **135b**. Referring to FIG. 4, the user device **110** may receive a user input modifying a device group, and the user device **110** may send the user input to the device control application **450a** (**402**), which may correspond to software operating with respect to the user device **110**. The user input may be received after the smart home supporting device(s) **130** have synchronized device group data with at least the device control supporting device(s) **135a** as described above in relation to FIG. 1. Such user input may modify the device group by adding a device to an existing device group, by removing a device from an existing device group, by modifying a group name of an existing device group, by modifying a device name for a device in an existing device group, or by creating a new device group. In some cases, the user input may modify the device group by modifying a setting related to or an operation of a device in an existing device group. For example, the user **105** may change a dim level of lights or may change a time when a TV is to turn off.

[0072] The device control application **450a** may send a modify device group event (**404**) to the GSS **145** (via the inbound events component **155** for example), where the event may include a device group identifier used by the device control application **450a** to identify the device group the user input modifies. The event may also include the user input or an indication of the modification made (e.g., update group name, add/remove device, update device operation, etc.). The event may be sent as an API request.

[0073] The GSS **145** may send, to a device control authorization component **455**, a request to determine whether the device control application **450a** is authorized to send events (**406**), in particular, for example, device group modification events. As described herein, the device control supporting device(s) **135a** may request and receive permissions from the user **105** to allow for device group synchronizations with the smart home supporting device(s) **130**. The device control authorization component **455** may be included in the smart home supporting device(s) **130**, the supporting device(s) **120**, and/or the user device **110**. The device control authorization component **455** may determine whether the device control application **450a** has the appropriate permissions by using data stored in the profile storage **670/770** or another data storage. The device control authorization component **455** may send, to the GSS component **145**, a response to the authorization check (**408**), which may indicate whether the device control application **450a** has the appropriate permissions or not (e.g., may be a Boolean value (True or False; 1 or 0; yes or no; etc.).

[0074] Assuming, in this example, that the device control application **450a** has the appropriate

permissions, the GSS **145** may retrieve, from the device group storage **165**, current device group data (**410**). The GSS **145** may use the device group identifier included in the modify event (**404**) to identify the current device group data. As described in relation to FIG. **3**, the device group storage **165** may include a device group identifier used by other device control applications, so that the device group identifier in a modify event from the device control application **450a** can be used to identify the corresponding device group data in the device group storage **165**.

[0075] The GSS **145** may modify the retrieved device group data based on the user input (**412**). The GSS **145** may store, in the device group storage **165**, the modified device group data (**414**). The modified device group data may be associated with the same/existing device group identifier. The modified device group data may reflect the modifications made by the user input, for example, addition/removal of a device. In case the user input creates a new device group, then the modified device group data may be associated with a new device group identifier, and may also include the device group identifier provided by the device control application **450a**.

[0076] In some embodiments, the GSS **145** may send the modified device group data to the device control supporting device(s) **135a**. In other embodiments, the modified device group data may not be communicated to the device control supporting device(s) **135a** as the request to modify the device group data originated from its device control application **450a** (the event in step **404**). In such embodiments, the GSS **145** may send a message (response to the event) to the device control supporting device(s) **135** that the request to modify group data has been completed.

[0077] In some cases, the GSS **145** may send the modified device group data to the device control supporting device(s) **135b** (**416**). The GSS **145** may do so when the modified device group includes a smart home device(s) **112** that is controlled using the device control supporting device(s) **135b**, whether or not that smart home device(s) is modified by the user input. In the case of a new group creation, the GSS **145** may send the new device group data when it includes a smart home device(s) controlled by the device control supporting device(s) **135b**.

[0078] The device control authorization component **455**, in some embodiments, may be configured to manage and check permissions for operations that device control applications can perform with respect to a user profile. In one example, the device control authorization component **455** may allow device control supporting device(s) **135** to request permission from the user **105** to synchronize user's device group data. In another example, the device control authorization component **455** may allow the device control supporting device(s) **135** to only view and edit device groups that contain smart home devices controlled by the device control supporting device(s) **135**. In yet another example, the device control authorization component **455** may allow the device control supporting device(s) **135** to only view and edit device locations for smart home devices controlled by the device control supporting device(s) **135**.

[0079] FIG. **5** is a signal flow diagram illustrating a process for modifying device group data based on a user input received at the device control application **140** of the smart home supporting device(s) **130**, according to embodiments of the present disclosure. Similar to as described above in relation to FIG. **4**, the user device **110** may receive a user input modifying a device group. In this example, the user input may be received via the device control application **140** (**502**).

[0080] The device control application **140** may send a modify device group event to the GSS **145** (**504**), and the event may include a device group identifier associated with the device group being modified by the user input. The device group identifier may be used by the smart home supporting device(s) **130** to identify the corresponding device group. The GSS **145** may retrieve current device group data from the device group storage **165** (**506**). The GSS **145** may retrieve the current device group data associated with the device group identifier included in the event.

[0081] The GSS **145** may modify the device group data based on the user input (**508**) (e.g., update group name, add/remove devices, etc.). The GSS **145** may determine other device control systems to send the modified device group data (**510**), where such determination may be based on a smart home device(s), included in the modified device group data, being controlled by the other device

control system. In this example, assuming the modified device group data includes a device **110**, **112** that is controlled by the device control supporting device(s) **135a**. The GSS **145** may store the modified device group data in the device group storage **165** (**512**), and the GSS **145** may send the modified device group data to the device control supporting device(s) **135a** (**514**).

[0082] Referring to FIGS. **4** and **5**, in some embodiments, when the modified device group data is sent, the GSS **145** may also request the device control supporting device(s) **135** to send updated data for the modified device group. For example, in case a device **110**, **112** is added to an existing group, the GSS **145** may request a device identifier used by the device control supporting device(s) **135** to identify the added device **110**, **112**. As another example, in a case a new device group is created, the GSS **145** may request a device group identifier used by the device control supporting device(s) **135** to identify the new device group. The data provided by the device control supporting device(s) **135** may be stored in the modified device group data in the device group storage **165**.

[0083] In some embodiments, communications from the GSS **145** to the device control supporting device(s) **135a** may be sent via the outbound events component **160**, and communications to the GSS **145** may be received via the inbound events component **155**.

[0084] In some embodiments, the smart home supporting device(s) **130** is configured to send portions of device group data associated with the particular device control supporting device(s) **135** to the respective device control supporting device(s) **135**. This may be done to limit exposure of data of a device control system to other device control systems. For example, a merged device group may include a first device controllable by the smart home supporting device(s) **130**, a second device controllable by the device control supporting device(s) **135a**, and a third device controllable by the device control supporting device(s) **135b**. In communicating merged device group data (e.g., steps **9a**, **9b** of FIG. **1**, step **416** of FIG. **4**, step **514** of FIG. **5**, etc.), the smart home supporting device(s) **130** may send, to the device control supporting device(s) **135a**, a first portion of the merged device group data including a group name, a group identifier, a device name and device identifier for the second device, based on the second device being controllable by the device control supporting device(s) **135a**. Continuing with the example, the smart home supporting device(s) **130** may send, to the device control supporting device(s) **135b**, a second portion of the merged device group data including a group name, a group identifier, a device name and device identifier for the third device, based on the third device being controllable by the device control supporting device(s) **135b**.

[0085] Referring now to FIG. **6**, the following describes example components, of a supporting device(s) **120** that may be used to process a user input. The user **105** may speak an input, and the user device **110** may receive audio **11** representing the spoken user input. For example, the user **105** may say “Alexa, what is the weather” or “Alexa, book me a plane ticket to Seattle.” In other examples, the user **105** may provide another type of input (e.g., selection of a button, selection of one or more displayed graphical interface elements, perform a gesture, etc.). The user device **110** may send input data to the supporting device(s) **120** for processing. In examples where the user input is a spoken user input, the input data may be audio data **611**. In other examples, the input data may be text data, or image data.

[0086] In the example of a spoken user input, a microphone or array of microphones (of or otherwise associated with the user device **110**) may continuously capture the audio **11**, and the user device **110** may continually process audio data, representing the audio **11**, as it is continuously captured, to determine whether speech is detected. The user device **110** may use various techniques to determine whether audio data includes speech. In some examples, the user device **110** may apply voice activity detection (VAD) techniques. Such techniques may determine whether speech is present in audio data based on various quantitative aspects of the audio data, such as the spectral slope between one or more frames of the audio data, the energy levels of the audio data in one or more spectral bands, the signal-to-noise ratios of the audio data in one or more spectral bands, or other quantitative aspects. In other examples, the user device **110** may implement a classifier

configured to distinguish speech from background noise. The classifier may be implemented by techniques such as linear classifiers, support vector machines, and decision trees. In still other examples, the user device **110** may apply Hidden Markov Model (HMM) or Gaussian Mixture Model (GMM) techniques to compare the audio data to one or more acoustic models in storage, which acoustic models may include models corresponding to speech, noise (e.g., environmental noise or background noise), or silence. Still other techniques may be used to determine whether speech is present in audio data.

[0087] Once speech is detected in the audio data representing the audio **11**, the user device **110** may determine if the speech is directed at the user device **110**. In some embodiments, such determination may be made using a wakeword detection component. The wakeword detection component may be configured to detect various wakewords. In at least some examples, each wakeword may correspond to a name of a different digital assistant. An example wakeword/digital assistant name is “Alexa.”

[0088] Wakeword detection is typically performed without performing linguistic analysis, textual analysis, or semantic analysis. Instead, the audio data, representing the audio **11**, is analyzed to determine if specific characteristics of the audio data match preconfigured acoustic waveforms, audio signatures, or other data corresponding to a wakeword.

[0089] Thus, the wakeword detection component may compare the audio data to stored data to detect a wakeword. One approach for wakeword detection applies general large vocabulary continuous speech recognition (LVCSR) systems to decode audio signals, with wakeword searching being conducted in the resulting lattices or confusion networks. Another approach for wakeword detection builds HMMs for each wakeword and non-wakeword speech signals, respectively. The non-wakeword speech includes other spoken words, background noise, etc. There can be one or more HMMs built to model the non-wakeword speech characteristics, which are named filler models. Viterbi decoding is used to search the best path in the decoding graph, and the decoding output is further processed to make the decision on wakeword presence. This approach can be extended to include discriminative information by incorporating a hybrid DNN-HMM decoding framework. In another example, the wakeword detection component **620** may be built on deep neural network (DNN)/recursive neural network (RNN) structures directly, without HMM being involved. Such an architecture may estimate the posteriors of wakewords with context data, either by stacking frames within a context window for DNN, or using RNN. Follow-on posterior threshold tuning or smoothing is applied for decision making. Other techniques for wakeword detection, such as those known in the art, may also be used.

[0090] Once the wakeword detection component detects a wakeword, the user device **110** may “wake” and send, to the supporting device(s) **120**, the input audio data **611** representing the spoken user input.

[0091] The supporting device(s) **120** may include an orchestrator component **630** configured to, among other things, coordinate data transmissions between components of the supporting device(s) **120**. The orchestrator component **630** may receive the audio data **611** from the user device **110**, and send the audio data **611** to an ASR component **650**.

[0092] The ASR component **650** transcribes the audio data **611** into ASR output data including one or more ASR hypotheses. An ASR hypothesis may be configured as a textual interpretation of the speech in the audio data **611**, or may be configured in another manner, such as one or more tokens. Each ASR hypothesis may represent a different likely interpretation of the speech in the audio data **611**. Each ASR hypothesis may be associated with a score (e.g., confidence score, probability score, or the like) representing the associated ASR hypothesis correctly represents the speech in the audio data **611**.

[0093] The ASR component **650** interprets the speech in the audio data **611** based on a similarity between the audio data **611** and pre-established language models. For example, the ASR component **650** may compare the audio data **611** with models for sounds (e.g., subword units, such

as phonemes, etc.) and sequences of sounds to identify words that match the sequence of sounds of the speech represented in the audio data **611**.

[0094] In at least some instances, instead of the user device **110** receiving a spoken natural language input, the user device **110** may receive a textual (e.g., typed) natural language input. The user device **110** may determine text data representing the textual natural language input, and may send the text data to the supporting device(s) **120**, wherein the text data is received by the orchestrator component **630**. The orchestrator component **630** may send the text data or ASR output data, depending on the type of natural language input received, to a NLU component **660**.

[0095] The NLU component **660** processes the ASR output data or text data to determine one or more NLU hypotheses embodied in NLU output data. The NLU component **660** may perform intent classification (IC) processing on the ASR output data or text data to determine an intent of the natural language input. An intent corresponds to an action to be performed that is responsive to the natural language input. To perform IC processing, the NLU component **660** may communicate with a database of words linked to intents. For example, a music intent database may link words and phrases such as “quiet,” “volume off,” and “mute” to a <Mute> intent. The NLU component **660** identifies intents by comparing words and phrases in ASR output data or text data to the words and phrases in an intents database. In some embodiments, the NLU component **660** may communicate with multiple intents databases, with each intents database corresponding to one or more intents associated with a particular skill.

[0096] For example, IC processing of the natural language input “play my workout playlist” may determine an intent of <PlayMusic>. For further example, IC processing of the natural language input “call mom” may determine an intent of <Call>. In another example, IC processing of the natural language input “call mom using video” may determine an intent of <VideoCall>. In yet another example, IC processing of the natural language input “what is today's weather” may determine an intent of <OutputWeather>.

[0097] The NLU component **660** may also perform named entity recognition (NER) processing on the ASR output data or text data to determine one or more portions, sometimes referred to as slots, of the natural language input that may be needed for post-NLU processing (e.g., processing performed by a skill). For example, NER processing of the natural language input “play [song name]” may determine an entity type of “SongName” and an entity value corresponding to the indicated song name. For further example, NER processing of the natural language input “call mom” may determine an entity type of “Recipient” and an entity value corresponding to “mom.” In another example, NER processing of the natural language input “what is today's weather” may determine an entity type of “Date” and an entity value of “today.”

[0098] In at least some embodiments, the intents identifiable by the NLU component **660** may be linked to one or more grammar frameworks with entity types to be populated with entity values. Each entity type of a grammar framework corresponds to a portion of ASR output data or text data that the NLU component **660** believes corresponds to an entity value. For example, a grammar framework corresponding to a <PlayMusic> intent may correspond to sentence structures such as “Play {Artist Name},” “Play {Album Name},” “Play {Song name},” “Play {Song name} by {Artist Name},” etc.

[0099] For example, the NLU component **660** may perform NER processing to identify words in ASR output data or text data as subject, object, verb, preposition, etc. based on grammar rules and/or models. Then, the NLU component **660** may perform IC processing using the identified verb to identify an intent. Thereafter, the NLU component **660** may again perform NER processing to determine a grammar model associated with the identified intent. For example, a grammar model for a <PlayMusic> intent may specify a list of entity types applicable to play the identified “object” and any object modifier (e.g., a prepositional phrase), such as {Artist Name}, {Album Name}, {Song name}, etc. The NER processing may then involve searching corresponding fields in a lexicon, attempting to match words and phrases in the ASR output data that NER processing

previously tagged as a grammatical object or object modifier with those identified in the lexicon.

[0100] NER processing may include semantic tagging, which is the labeling of a word or combination of words according to their type/semantic meaning. NER processing may include parsing ASR output data or text data using heuristic grammar rules, or a model may be constructed using techniques such as hidden Markov models, maximum entropy models, log linear models, conditional random fields (CRFs), and the like. For example, NER processing with respect to a music skill may include parsing and tagging ASR output data or text data corresponding to “play mother's little helper by the rolling stones” as {Verb}: “Play,” {Object}: “mother's little helper,” {Object Preposition}: “by,” and {Object Modifier}: “the rolling stones.” The NER processing may identify “Play” as a verb based on a word database associated with the music skill, which IC processing determines corresponds to a <PlayMusic> intent.

[0101] The NLU component **660** may generate NLU output data including one or more NLU hypotheses, with each NLU hypothesis including an intent and optionally one or more entity types and corresponding entity values. In some embodiments, the NLU component **660** may perform IC processing and NER processing with respect to different skills. One skill may support the same or different intents than another skill. Thus, the NLU output data may include multiple NLU hypotheses, with each NLU hypothesis corresponding to IC processing and NER processing performed on the ASR output or text data with respect to a different skill.

[0102] The skill shortlisting component **665** is configured to determine a subset of skill components, implemented by or in communication with the supporting device(s) **120**, that may perform an action responsive to the (spoken) user input. Without the skill shortlisting component **665**, the NLU component **660** may process ASR output data input thereto with respect to every skill component of or in communication with the supporting device(s) **120**. By implementing the skill shortlisting component **665**, the NLU component **660** may process ASR output data with respect to only the skill components the skill shortlisting component **665** determines are likely to execute with respect to the user input. This reduces total compute power and latency attributed to NLU processing.

[0103] The skill shortlisting component **665** may include one or more ML models. The ML model(s) may be trained to recognize various forms of user inputs that may be received by the supporting device(s) **120**. For example, during a training period, a skill component developer may provide training data representing sample user inputs that may be provided by a user to invoke the skill component. For example, for a ride sharing skill component, a skill component developer may provide training data corresponding to “get me a cab to [location],” “get me a ride to [location],” “book me a cab to [location],” “book me a ride to [location],” etc.

[0104] The supporting device(s) **120** may use the sample user inputs, provided by a skill component developer, to determine other potentially related user input structures that users may try to use to invoke the particular skill component. The ML model(s) may be further trained using these potentially related user input structures. During training, the skill component developer may be queried regarding whether the determined other user input structures are permissible, from the perspective of the skill component developer, to be used to invoke the skill component. The potentially related user input structures may be derived by one or more ML models, and may be based on user input structures provided by different skill component developers.

[0105] The skill component developer may also provide training data indicating grammar and annotations.

[0106] Each ML model, of the skill shortlisting component **665**, may be trained with respect to a different skill component. Alternatively, the skill shortlisting component **665** may implement one ML model per domain, such as one ML model for skill components associated with a weather domain, one ML model for skill components associated with a ride sharing domain, etc.

[0107] The sample user inputs provided by a skill component developer, and potentially related sample user inputs determined by the supporting device(s) **120**, may be used as binary examples to

train a ML model associated with a skill component. For example, some sample user inputs may be positive examples (e.g., user inputs that may be used to invoke the skill component). Other sample user inputs may be negative examples (e.g., user inputs that may not be used to invoke the skill component).

[0108] As described above, the skill shortlisting component **665** may include a different ML model for each skill component, a different ML model for each domain, or some other combination of ML models. In some embodiments, the skill shortlisting component **665** may alternatively include a single ML model. This ML model may include a portion trained with respect to characteristics (e.g., semantic characteristics) shared by all skill components. The ML model may also include skill component-specific portions, with each skill component-specific portion being trained with respect to a specific skill component. Implementing a single ML model with skill component-specific portions may result in less latency than implementing a different ML model for each skill component because the single ML model with skill component-specific portions limits the number of characteristics processed on a per skill component level.

[0109] The portion, trained with respect to characteristics shared by more than one skill component, may be clustered based on domain. For example, a first portion, of the portion trained with respect to multiple skill components, may be trained with respect to weather domain skill components; a second portion, of the portion trained with respect to multiple skill components, may be trained with respect to music domain skill components; a third portion, of the portion trained with respect to multiple skill components, may be trained with respect to travel domain skill components; etc.

[0110] The skill shortlisting component **665** may make binary (e.g., yes or no) determinations regarding which skill components relate to the ASR output data. The skill shortlisting component **665** may make such determinations using the one or more ML models described herein above. If the skill shortlisting component **665** implements a different ML model for each skill component, the skill shortlisting component **665** may run the ML models that are associated with enabled skill components as indicated in a user profile associated with the user device **110** and/or the user **105**.

[0111] The skill shortlisting component **665** may generate an n-best list of skill components that may execute with respect to the user input represented in the ASR output data. The size of the n-best list of skill components is configurable. In an example, the n-best list of skill components may indicate every skill component of, or in communication with, the supporting device(s) **120** as well as contain an indication, for each skill component, representing whether the skill component is likely to execute the user input represented in the ASR output data. In another example, instead of indicating every skill component, the n-best list of skill components may only indicate the skill components that are likely to execute the user input represented in the ASR output data. In yet another example, the skill shortlisting component **665** may implement thresholding such that the n-best list of skill components may indicate no more than a maximum number of skill components. In another example, the skill components included in the n-best list of skill components may be limited by a threshold score, where only skill components associated with a likelihood to handle the user input above a certain score are included in the n-best list of skill components.

[0112] The ASR output data may correspond to more than one ASR hypothesis. When this occurs, the skill shortlisting component **665** may output a different n-best list of skill components for each ASR hypothesis. Alternatively, the skill shortlisting component **665** may output a single n-best list of skill components representing the skill components that are related to the multiple ASR hypotheses represented in the ASR output data.

[0113] As indicated above, the skill shortlisting component **665** may implement thresholding such that an n-best list of skill components output therefrom may include no more than a threshold number of entries. If the ASR output data includes more than one ASR hypothesis, the n-best list of skill components may include no more than a threshold number of entries irrespective of the number of ASR hypotheses output by the ASR component **650**. Additionally or alternatively, the n-

best list of skill components may include no more than a threshold number of entries for each ASR hypothesis (e.g., no more than five entries for a first ASR hypothesis, no more than five entries for a second ASR hypothesis, etc.).

[0114] Additionally or alternatively to making a binary determination regarding whether a skill component potentially relates to the ASR output data, the skill shortlisting component **665** may generate confidence scores representing likelihoods that skill components relate to the ASR output data. The skill shortlisting component **665** may perform matrix vector modification to obtain confidence scores for all skill components in a single instance of processing of the ASR output data.

[0115] An n-best list of skill components including confidence scores that may be output by the skill shortlisting component **665** may be represented as, for example: [0116] Story skill component, 0.67 [0117] Recipe skill component, 0.62 [0118] Information skill component, 0.57 [0119] Shopping skill component, 0.62

[0120] As indicated, the confidence scores output by the skill shortlisting component **665** may be numeric values. The confidence scores output by the skill shortlisting component **665** may alternatively be binned values (e.g., high, medium, low).

[0121] The n-best list of skill components may only include entries for skill components having a confidence score satisfying (e.g., meeting or exceeding) a minimum threshold confidence score. Alternatively, the skill shortlisting component **665** may include entries for all skill components associated with enabled skill components of the current user, even if one or more of the skill components are associated with confidence scores that do not satisfy the minimum threshold confidence score.

[0122] The skill shortlisting component **665** may consider other data when determining which skill components may relate to the user input represented in the ASR output data as well as respective confidence scores. The other data may include usage history data, data indicating the skill components that are enabled with respect to the user device **110** and/or user **105**, data indicating a device type of the user device **110**, data indicating a speed of the user device **110**, a location of the user device **110**, data indicating a skill component that was being used to output content via the user device **110** when the user device **110** received the instant user input, etc.

[0123] The thresholding implemented with respect to the n-best list of skill components generated by the skill shortlisting component **665** as well as the different types of other data considered by the skill shortlisting component **665** are configurable.

[0124] As described above, the supporting device(s) **120** may perform speech processing using two different components (e.g., the ASR component **650** and the NLU component **660**). In at least some embodiments, the supporting device(s) **120** may implement a spoken language understanding (SLU) component **660** configured to process audio data **611** to determine NLU output data.

[0125] The SLU component **640** may be equivalent to a combination of the ASR component **650** and the NLU component **660**. Yet, the SLU component **640** may process audio data **611** and directly determine the NLU output data, without an intermediate step of generating ASR output data. As such, the SLU component **640** may take audio data **611** representing a spoken natural language input and attempt to make a semantic interpretation of the spoken natural language input. That is, the SLU component **640** may determine a meaning associated with the spoken natural language input and then implement that meaning. For example, the SLU component **640** may interpret audio data **611** representing a spoken natural language input in order to derive a desired action. The SLU component **640** may output a most likely NLU hypothesis, or multiple NLU hypotheses associated with respective confidence or other scores (such as probability scores, etc.).

[0126] The supporting device(s) **120** may include a gesture detection component (not illustrated in FIG. **6**). The supporting device(s) **120** may receive image data representing a gesture, and the gesture detection component may process the image data to determine a gesture represented therein. The gesture detection component may implement art-/industry-known gesture detection

processes.

[0127] In embodiments where the supporting device(s) **120** receives non-image data (e.g., text data) representing a gesture, the orchestrator component **630** may be configured to determine what downstream processing is to be performed in response to the gesture.

[0128] The system may include a skill selection component **685** is configured to determine a skill component, or n-best list of skill components each associated with a confidence score/value, to execute to respond to the user input. The skill selection component **685** may include a skill component proposal component, a skill component pre-response component, and a skill component ranking component.

[0129] The skill component proposal component is configured to determine skill components capable of processing in response to the user input. In addition to receiving the NLU output data, the skill component proposal component may receive context data corresponding to the user input. For example, the context data may indicate a skill component that was causing the user device **110** to output content (e.g., music, video, synthesized speech, etc.) when the user device **110** captured the user input, one or more skill components that are indicated as enabled in a profile (as stored in the profile storage **670**) associated with the user **105**, output capabilities of the user device **110**, a geographic location of the user device **110**, and/or other context data corresponding to the user input.

[0130] The skill component proposal component may implement skill component proposal rules. A skill component developer, via a skill component developer device, may provide one or more rules representing when a skill component should be invoked to respond to a user input. In some embodiments, such a rule may be specific to an intent. In such embodiments, if a skill component is configured to execute with respect to multiple intents, the skill component may be associated with more than one rule (e.g., each rule corresponding to a different intent capable of being handled by the skill component). In addition to being specific to an intent, a rule may indicate one or more entity identifiers with respect to which the skill component should be invoked. For further example, a rule may indicate output capabilities of a device, a geographic location, and/or other conditions.

[0131] Each skill component may be associated with each rule corresponding to the skill component. As an example, a rule may indicate a video skill component may execute when a user input corresponds to a “PlayVideo” intent and the device includes or is otherwise associated with a display. As another example, a rule may indicate a music skill component may execute when a user input corresponds to a “PlayMusic” intent and music is being output by a device when the device captures the user input. It will be appreciated that other examples are possible. The foregoing rules enable skill components to be differentially proposed at runtime, based on various conditions, in systems where multiple skill components are configured to execute with respect to the same intent.

[0132] The skill component proposal component, using the NLU output data, received context data, and the foregoing described skill component proposal rules, determines skill components configured to process in response to the user input. Thus, in some embodiments, the skill component proposal component may be implemented as a rules engine. In some embodiments, the skill component proposal component may make binary (e.g., yes/no, true/false, etc.) determinations regarding whether a skill component is configured to process in response to the user input. For example, the skill component proposal component may determine a skill component is configured to process, in response to the user input, if the skill component is associated with a rule corresponding to the intent, represented in the NLU output data, and the context data.

[0133] In some embodiments, the skill component proposal component may make such binary determinations with respect to all skill components. In some embodiments, the skill component proposal component may make the binary determinations with respect to only some skill components (e.g., only skill components indicated as enabled in the user profile of the user **105**).

[0134] After the skill component proposal component is finished processing, the skill component pre-response component may be called to execute. The skill component pre-response component is

configured to query skill components, determined by the skill component proposal component as configured to process the user input, as to whether the skill components are in fact able to respond to the user input. The skill component pre-response component may take as input the NLU output data including one or more NLU hypotheses, where each of the one or more NLU hypotheses is associated with a particular skill component determined by the skill component proposal component as being configured to respond to the user input.

[0135] The skill component pre-response component sends a pre-response query to each skill component determined by the skill component proposal component. A pre-response query may include the NLU hypothesis associated with the skill component, and optionally other context data corresponding to the user input.

[0136] A skill component may determine, based on a received pre-response query and optionally other data available to the skill component, whether the skill component is capable of responding to the user input. For example, a skill component may generate a pre-response indicating the skill component can respond to the user input, indicating the skill component needs more data to determine whether the skill component can respond to the user input, or indicating the skill component cannot respond to the user input.

[0137] In situations where a skill component's pre-response indicates the skill component can respond to the user input, or indicating the skill component needs more information, the skill component's pre-response may also include various other data representing a strength of the skill component's potential response to the user input. Such other data may positively influence the skill component's ranking by the skill component ranking component of the skill selection component **685**. For example, such other data may indicate capabilities (e.g., output capabilities or components such as a connected screen, loudspeaker, etc.) of a device to be used to output the skill component's response; pricing data corresponding to a product or service the user input is requesting be purchased or is requesting information for; availability of a product the user input is requesting be purchased; whether there are shipping fees for a product the user input is requesting be purchased; whether the user **105** already has a profile and/or subscription with the skill component; that the user **105** does not have a subscription with the skill component, but that there is a free trial/tier the skill component is offering; with respect to a taxi skill component, a cost of a trip based on start and end locations, how long the user **105** would have to wait to be picked up, etc.; and/or other data available to the skill component that is related to the skill component's processing of the user input. In some embodiments, a skill component's pre-response may include an indicator (e.g., flag, representing a strength of the skill component's ability to personalize its response to the user input).

[0138] In some embodiments, a skill component's pre-response may be configured to a pre-defined schema. By requiring pre-responses to conform to a specific schema (e.g., by requiring skill components to only be able to provide certain types of data in pre-responses), new skill components may be onboarded into the skill component selection functionality without needing to reconfigure the skill selection component **685** each time a new skill component is onboarded. Moreover, requiring pre-responses to conform to a schema limits the amount of values needed to be used to train and implement a ML model for ranking skill components.

[0139] In some embodiments, a skill component's pre-response may indicate whether the skill component requests exclusive display access (i.e., whether the skill component requests its visual data be presented on an entirety of the display).

[0140] After the skill component pre-response component queries the skill components for pre-responses, the skill component ranking component may be called to execute. The skill component ranking component may be configured to select a single skill component, from among the skill components determined by the skill component proposal component, to respond to the user input. In some embodiments, the skill component ranking component may implement a ML model. In some embodiments, the ML model may be a deep neural network (DNN).

[0141] The skill component ranking component may take as input the NLU output data, the skill

component pre-responses, one or more skill component preferences of the user **105** (e.g., as represented in a user profile or group profile stored in the profile storage **670**), NLU confidence scores of the NLU output data, a device type of the user device **110**, data indicating whether the user device **110** was outputting content when the user input was received, and/or other context data available to the skill component ranking component.

[0142] The skill component ranking component ranks the skill components using the ML model. Things that may increase a skill component's ranking include, for example, that the skill component is associated with a pre-response indicating the skill component can generate a response that is personalized to the user **105**, that a NLU hypothesis corresponding to the skill component is associated with a NLU confidence score satisfying a condition (e.g., a threshold NLU confidence score) that the skill component was outputting content via the user device **110** when the user device **110** received the user input, etc. Things that may decrease a skill component's ranking include, for example, that the skill component is associated with a pre-response indicating the skill component cannot generate a response that is personalized to the user **105**, that a NLU hypothesis corresponding to the skill component is associated with a NLU confidence score failing to satisfy a condition (e.g., a threshold NLU confidence score, etc.).

[0143] The skill component ranking component may generate a score for each skill component determined by the skill component proposal component, where the score represents a strength with which the skill component ranking component recommends the associated skill component be executed to respond to the user input. Such a confidence score may be a numeric score (e.g., between 0 and 1) or a binned score (e.g., low, medium, high).

[0144] The supporting device(s) **120** may include or otherwise communicate with one or more skill components **690**. A skill component **690** may process NLU output data and perform one or more actions in response thereto. For example, for NLU output data including a <PlayMusic> intent, an “artist” entity type, and an artist name as an entity value, a music skill component may output music sung by the indicated artist. For further example, for NLU output data including a <TurnOn> intent, a “device” entity type, and an entity value of “lights,” a smart home skill component may cause one or more “smart” lights to operate in an “on” state. In another example, for NLU output data including an <Output Weather> intent, a “location” entity type, and an entity value corresponding to a geographic location of the user device **110**, a weather skill component may output weather information for the geographic location. For further example, for NLU output data including a <BookRide> intent, a taxi skill component may book a requested ride. In another example, for NLU output data including a <BuyPizza> intent, a restaurant skill component may place an order for a pizza. In another example, for NLU output data including an <OutputStory> intent and a “title” entity type and corresponding title entity value, a story skill component may output a story corresponding to the title.

[0145] A skill supporting device(s) **625** may communicate with the skill component(s) **690** within the supporting device(s) **120** directly and/or via the orchestrator component **630**. A skill supporting device(s) **625** may be configured to perform one or more actions. A skill component **690** may enable a skill supporting device(s) **625** to execute specific functionality in order to provide data or perform some other action requested by a user. For example, a weather service skill may enable a skill supporting device(s) **625** to provide weather information to the supporting device(s) **120**, a car service skill may enable a skill supporting device(s) **625** to book a trip using a taxi or ride sharing service, an order pizza skill may enable a skill supporting device(s) **625** to order a pizza using a restaurant's online ordering system, etc. Additional types of skills include home automation skills (e.g., skills that enable a user to control home devices such as lights, door locks, cameras, thermostats, etc.), entertainment device skills (e.g., skills that enable a user to control entertainment devices such as smart televisions), video skills, flash briefing skills, as well as custom skills that are not associated with any pre-configured type of skill. The supporting device(s) **120** may include a skill component **690** dedicated to interacting with the skill supporting device(s) **625**. A skill, skill

device, or skill component may include a skill component **690** operated by the supporting device(s) **120** and/or operated by the skill supporting device(s) **625**.

[0146] A skill component may operate in conjunction between the user device **110**, the supporting device(s) **120** and/or other devices, such as a restaurant electronic ordering system, a taxi electronic booking system, etc. in order to complete certain functions. Inputs to a skill component **690** may come from speech processing interactions or through other interactions or input sources.

[0147] A skill component **690**/skill supporting device(s) **625** may be associated with a domain, a non-limiting list of which includes a smart home domain, a music domain, a video domain, a weather domain, a communications domain, a flash briefing domain, a shopping domain, and a custom domain.

[0148] The skill component **690** may process to determine output data responsive to the spoken user input (e.g., based on the intent and entity data as represented in the NLU output data received by the skill component **690**).

[0149] The supporting device(s) **120** may include a TTS component **680** that generates audio data including synthesized speech. The TTS component **680** is configured to generate output audio data including synthesized speech. The TTS component **680** may perform speech synthesis using one or more different methods. In one method of synthesis called unit selection, the TTS component **680** matches a database of recorded speech against the data input to the TTS component **680**. The TTS component **680** matches the input data against spoken audio units in the database. Matching units are selected and concatenated together to form a speech output. Each unit includes an audio waveform corresponding with a phonetic unit, such as a short .wav file of the specific sound, along with a description of the various acoustic features associated with the .wav file, such as its pitch, energy, etc., as well as other information, such as where the phonetic unit appears in a word, sentence, or phrase, the neighboring phonetic units, etc. Using all the information in the unit database, the TTS component **680** may match units to the input data to create a natural sounding waveform. The unit database may include multiple examples of phonetic units to provide the TTS component **680** with many different options for concatenating units into speech. One benefit of unit selection is that, depending on the size of the database, a natural sounding speech output may be generated. The larger the unit database, the more likely the TTS component **680** will be able to construct natural sounding speech.

[0150] Unit selection speech synthesis may be performed as follows. Unit selection includes a two-step process. First the TTS component **680** determines what speech units to use and then it combines them so that the particular combined units match the desired phonemes and acoustic features to create the desired speech output. Units may be selected based on a cost function which represents how well particular units fit the speech segments to be synthesized. The cost function may represent a combination of different costs representing different aspects of how well a particular speech unit may work for a particular speech segment. For example, a target cost indicates how well a given speech unit matches the features of a desired speech output (e.g., pitch, prosody, etc.). A join cost represents how well a speech unit matches a consecutive speech unit for purposes of concatenating the speech units together in the eventual synthesized speech. The overall cost function is a combination of target cost, join cost, and other costs that may be determined by the TTS component **680**. As part of unit selection, the TTS component **680** chooses the speech unit with the lowest overall combined cost. For example, a speech unit with a very low target cost may not necessarily be selected if its join cost is high.

[0151] In another method of synthesis called parametric synthesis, parameters such as frequency, volume, noise, etc. are varied by the TTS component **680** to create an artificial speech waveform output. Parametric synthesis may use an acoustic model and various statistical techniques to match data, input to the TTS component **680**, with desired output speech parameters. Parametric synthesis may include the ability to be accurate at high processing speeds, as well as the ability to process speech without large databases associated with unit selection, but also typically produces an output

speech quality that may not match that of unit selection. Unit selection and parametric techniques may be performed individually or combined together and/or combined with other synthesis techniques to produce speech audio output.

[0152] Parametric speech synthesis may be performed as follows. The TTS component **680** may include an acoustic model, or other models, which may convert data, input to the TTS component **680**, into a synthetic acoustic waveform based on audio signal manipulation. The acoustic model includes rules that may be used to assign specific audio waveform parameters to input phonetic units and/or prosodic annotations. The rules may be used to calculate a score representing a likelihood that a particular audio output parameter(s), such as frequency, volume, etc., corresponds to the portion of the input data.

[0153] The TTS component **680** may use a number of techniques to match speech to be synthesized with input phonetic units and/or prosodic annotations. One common technique is using Hidden Markov Models (HMMs). HMMs may be used to determine probabilities that audio output should match textual input. HMMs may be used to translate from parameters from the linguistic and acoustic space to the parameters to be used by a vocoder (i.e., a digital voice encoder) to artificially synthesize the desired speech. Using HMMs, a number of states are presented, in which the states together represent one or more potential acoustic parameters to be output to the vocoder and each state is associated with a model, such as a Gaussian mixture model. Transitions between states may also have an associated probability, representing a likelihood that a current state may be reached from a previous state. Sounds to be output may be represented as paths between states of the HMM and multiple paths may represent multiple possible audio matches for the same input text. Each portion of text may be represented by multiple potential states corresponding to different known pronunciations of phonemes and their parts, such as the phoneme identity, stress, accent, position, etc. An initial determination of a probability of a potential phoneme may be associated with one state. As new text is processed by the TTS component **680**, the state may change or stay the same, based on the processing of the new text. For example, the pronunciation of a previously processed word might change based on later processed words. A Viterbi algorithm may be used to find the most likely sequence of states based on the processed text. The HMMs may generate speech in parametrized form including parameters such as fundamental frequency (f0), noise envelope, spectral envelope, etc. that are translated by a vocoder into audio segments. The output parameters may be configured for particular vocoders such as a STRAIGHT vocoder, TANDEM-STRAIGHT vocoder, HNM (harmonic plus noise) based vocoders, CELP (code-excited linear prediction) vocoders, GlottHMM vocoders, HSM (harmonic/stochastic model) vocoders, or others.

[0154] In addition to calculating potential states for one audio waveform as a potential match to a phonetic unit, the TTS component **680** may also calculate potential states for other potential audio outputs, such as various ways of pronouncing phoneme/E/, as potential acoustic matches for the phonetic unit. In this manner multiple states and state transition probabilities may be calculated.

[0155] The probable states and probable state transitions calculated by the TTS component **680** may lead to a number of potential audio output sequences. Based on the acoustic model and other potential models, the potential audio output sequences may be scored according to a confidence level of the TTS component **680**. The highest scoring audio output sequence, including a stream of parameters to be synthesized, may be chosen and digital signal processing may be performed by a vocoder or similar component to create an audio output including synthesized speech waveforms corresponding to the parameters of the highest scoring audio output sequence and, if the proper sequence was selected, also corresponding to the input data.

[0156] The supporting device(s) **120** may include a user recognition component **695**. The user recognition component **695** may recognize one or more users using various data. The user recognition component **695** may take as input the audio data **611**. The user recognition component **695** may perform user recognition by comparing speech characteristics, in the audio data **611**, to stored speech characteristics of users. The user recognition component **695** may additionally or

alternatively perform user recognition by comparing biometric data (e.g., fingerprint data, iris data, retina data, etc.), received by the supporting device(s) **120** in correlation with a natural language input, to stored biometric data of users. The user recognition component **695** may additionally or alternatively perform user recognition by comparing image data (e.g., including a representation of at least a feature of a user), received by the supporting device(s) **120** in correlation with a natural language input, with stored image data including representations of features of different users. The user recognition component **695** may perform other or additional user recognition processes, including those known in the art. For a particular natural language input, the user recognition component **695** may perform processing with respect to stored data of users associated with the user device **110** that received the natural language input.

[0157] The user recognition component **695** determines whether a natural language input originated from a particular user. For example, the user recognition component **695** may determine a first value representing a likelihood that a natural language input originated from a first user, a second value representing a likelihood that the natural language input originated from a second user, etc. The user recognition component **695** may also determine an overall confidence regarding the accuracy of user recognition processing.

[0158] The user recognition component **695** may output a single user identifier corresponding to the most likely user that originated the natural language input. Alternatively, the user recognition component **695** may output multiple user identifiers (e.g., in the form of an N-best list) with respective values representing likelihoods of respective users originating the natural language input. The output of the user recognition component **695** may be used to inform NLU processing, processing performed by a skill component **690**, as well as processing performed by other components of the supporting device(s) **120** and/or other systems.

[0159] The supporting device(s) **120** may include profile storage **670**. The profile storage **670** may include a variety of data related to individual users, groups of users, devices, etc. As used herein, a “profile” refers to a set of data associated with a user, group of users, device, etc. The data of a profile may include preferences specific to the user, group of users, device, etc.; input and output capabilities of one or more devices; internet connectivity data; user bibliographic data; subscription data; skill component enablement data; and/or other data.

[0160] The profile storage **670** may include one or more user profiles. Each user profile may be associated with a different user identifier. Each user profile may include various user identifying data (e.g., name, gender, address, language(s), etc.). Each user profile may also include preferences of the user. Each user profile may include one or more device identifiers, each representing a respective device registered to the user. Each user profile may include skill component identifiers of skill components that the user has enabled. When a user enables a skill component, the user is providing permission to allow the skill component to execute with respect to the user's inputs. If a user does not enable a skill component, the skill component may be prevented from processing with respect to the user's inputs.

[0161] The profile storage **670** may include one or more group profiles. Each group profile may be associated with a different group identifier. A group profile may be specific to a group of users. That is, a group profile may be associated with two or more individual user profiles. For example, a group profile may be a household profile that is associated with user profiles associated with multiple users of a single household. A group profile may include preferences shared by all the user profiles associated therewith. Each user profile associated with a group profile may additionally include preferences specific to the user associated therewith. That is, a user profile may include preferences unique from one or more other user profiles associated with the same group profile. A user profile may be a stand-alone profile or may be associated with a group profile. A group profile may be associated with (or include) one or more device profiles corresponding to one or more devices associated with the group profile.

[0162] The profile storage **670** may include one or more device profiles. Each device profile may

be associated with a different device identifier. A device profile may include various device identifying data, input/output characteristics, networking characteristics, etc. A device profile may also include one or more user identifiers, corresponding to one or more user profiles associated with the device profile. For example, a household device's profile may include the user identifiers of users of the household.

[0163] The foregoing describes illustrative components and processing of the supporting device(s) **120**. The following describes illustrative components and processing of the user device **110**. As illustrated in FIG. 7, in at least some embodiments the supporting device(s) **120** may receive the audio data **611** from the user device **110**, to recognize speech in the received audio data **611**, and to perform functions in response to the recognized speech. In at least some embodiments, these functions involve sending directives (e.g., commands), from the supporting device(s) **120** to the user device **110** to cause the user device **110** to perform an action, such as output synthesized speech (responsive to the spoken user input) via a loudspeaker(s), and/or control one or more secondary devices by sending control commands to the one or more secondary devices.

[0164] Thus, when the user device **110** is able to communicate with the supporting device(s) **120** over the network(s) **199**, some or all of the functions capable of being performed by the supporting device(s) **120** may be performed by sending one or more directives over the network(s) **199** to the user device **110**, which, in turn, may process the directive(s) and perform one or more corresponding actions. For example, the supporting device(s) **120**, using a remote directive that is included in response data (e.g., a remote response), may instruct the user device **110** to output synthesized speech via a loudspeaker(s) of (or otherwise associated with) the user device **110**, to output content (e.g., music) via the loudspeaker(s) of (or otherwise associated with) the user device **110**, to display content on a display of (or otherwise associated with) the user device **110**, and/or to send a directive to a secondary device (e.g., a directive to turn on a smart light). It will be appreciated that the supporting device(s) **120** may be configured to provide other functions in addition to those discussed herein, such as, without limitation, providing step-by-step directions for navigating from an origin location to a destination location, conducting an electronic commerce transaction on behalf of the user **105** as part of a shopping function, establishing a communication session (e.g., an audio or video call) between the user **105** and another user, and so on.

[0165] As noted previously, the user device **110** may include a wakeword detection component **620** configured to detect a wakeword (e.g., "Alexa") that indicates to the user device **110** that the audio data **611** is to be processed for determining NLU output data. In at least some embodiments, a hybrid selector **724**, of the user device **110**, may send the audio data **611** to the wakeword detection component **620**. If the wakeword detection component **620** detects a wakeword in the audio data **611**, the wakeword detection component **620** may send an indication of such detection to the hybrid selector **724**. In response to receiving the indication, the hybrid selector **724** may send the audio data **611** to the supporting device(s) **120** and/or an ASR component **750** implemented by the device **110**. The wakeword detection component **620** may also send an indication, to the hybrid selector **724**, representing a wakeword was not detected. In response to receiving such an indication, the hybrid selector **724** may refrain from sending the audio data **611** to the supporting device(s) **120**, and may prevent the ASR component **750** from processing the audio data **611**. In this situation, the audio data **611** can be discarded.

[0166] The user device **110** may conduct its own speech processing using on-device language processing components (such as a SLU component **740**, the ASR component **750**, and/or a NLU component **760**) similar to the manner discussed above with respect to the system-implemented SLU component **640**, ASR component **650**, and NLU component **660**. The user device **110** may also internally include, or otherwise have access to, other components such as one or more skill components **790** (configured to process in a similar manner to the one or more skills components **690** implemented by and/or in communication with the supporting device(s) **120**/the skill supporting device(s) **625**), a profile storage **770** (configured to store similar profile data to the

profile storage **670** implemented by the supporting device(s) **120**), a TTS component **780** (configured to process in a similar manner to the TTS component **680** implemented by the supporting device(s) **120**), a skill selection component **785** (configured to process in a similar manner to the skill selection component **685** implemented by the supporting device(s) **120**), a skill shortlisting component **765** (configured to process in a similar manner to the skill shortlisting component **665** implemented by the supporting device(s) **120**), and/or other components. In at least some embodiments, the profile storage **770** may only store profile data for a user or group of users specifically associated with the user device **110**.

[0167] In at least some embodiments, the on-device language processing components may not have the same capabilities as the language processing components implemented by the supporting device(s) **120**. For example, the on-device language processing components may be configured to handle only a subset of the user inputs that may be handled by the system-implemented language processing components. For example, such subset of user inputs may correspond to local-type user inputs, such as those controlling devices or components associated with a user's home. In such circumstances the on-device language processing components may be able to more quickly interpret and respond to a local-type user input, for example, than processing that involves the supporting device(s) **120**. If the user device **110** attempts to process a user input for which the on-device language processing components are not necessarily best suited, the NLU output data, determined by the on-device components, may have a low confidence or other metric indicating that the processing by the on-device language processing components may not be as accurate as the processing done by the supporting device(s) **120**.

[0168] The hybrid selector **724**, of the user device **110**, may include a hybrid proxy (HP) **726** configured to proxy traffic to/from the supporting device(s) **120**. For example, the HP **726** may be configured to send messages to/from a hybrid execution controller (HEC) **727** of the hybrid selector **724**. For example, command/directive data received from the supporting device(s) **120** can be sent to the HEC **727** using the HP **726**. The HP **726** may also be configured to allow the audio data **611** to pass to the supporting device(s) **120** while also receiving (e.g., intercepting) this audio data **611** and sending the audio data **611** to the HEC **727**.

[0169] In at least some embodiments, the hybrid selector **724** may further include a local request orchestrator (LRO) **728** configured to notify the ASR component **750** about the availability of the audio data **611**, and to otherwise initiate the operations of on-device language processing when the audio data **611** becomes available. In general, the hybrid selector **724** may control execution of on-device language processing, such as by sending “execute” and “terminate” events/instructions. An “execute” event may instruct a component to continue any suspended execution (e.g., by instructing the component to execute on a previously-determined intent in order to determine a directive). Meanwhile, a “terminate” event may instruct a component to terminate further execution, such as when the user device **110** receives directive data from the supporting device(s) **120** and chooses to use that remotely-determined directive data.

[0170] Thus, when the audio data **611** is received, the HP **726** may allow the audio data **611** to pass through to the supporting device(s) **120** and the HP **726** may also input the audio data **611** to the ASR component **750** by routing the audio data **611** through the HEC **727** of the hybrid selector **724**, whereby the LRO **728** notifies the ASR component **750** of the audio data **611**. At this point, the hybrid selector **724** may wait for response data from either or both the supporting device(s) **120** and/or the on-device language processing components. However, the disclosure is not limited thereto, and in some examples the hybrid selector **724** may send the audio data **611** only to the ASR component **750** without departing from the disclosure. For example, the user device **110** may process the audio data **611** on-device without sending the audio data **611** to the supporting device(s) **120**.

[0171] The ASR component **750** is configured to receive the audio data **611** from the hybrid selector **724**, and to recognize speech in the audio data **611**, and the NLU component **760** is

configured to determine an intent from the recognized speech (an optionally one or more named entities), and to determine how to act on the intent by generating directive data (e.g., instructing a component to perform an action). In some cases, a directive may include a description of the intent (e.g., an intent to turn off {device A}). In some cases, a directive may include (e.g., encode) an identifier of a second device(s), such as kitchen lights, and an operation to be performed at the second device(s). Directive data may be formatted using Java, such as JavaScript syntax, or JavaScript-based syntax. This may include formatting the directive using JSON. In at least some embodiments, a device-determined directive may be serialized, much like how remotely-determined directives may be serialized for transmission in data packets over the network(s) **199**.

In at least some embodiments, a device-determined directive may be formatted as a programmatic application programming interface (API) call with a same logical operation as a remotely-determined directive. In other words, a device-determined directive may mimic a remotely-determined directive by using a same, or a similar, format as the remotely-determined directive. [0172] NLU output data (output by the NLU component **760**) may be selected as usable to respond to a user input, and local response data may be sent to the hybrid selector **724**, such as a “ReadyToExecute” response. The hybrid selector **724** may then determine whether to use directive data from the on-device components to respond to the user input, to use directive data received from the supporting device(s) **120**, assuming a remote response is even received (e.g., when the user device **110** is able to access the supporting device(s) **120** over the network(s) **199**), or to determine output data requesting additional information from the user **105**.

[0173] The user device **110** and/or the supporting device(s) **120** may associate a unique identifier with each user input. The user device **110** may include the unique identifier when sending the audio data **611** to the supporting device(s) **120**, and the response data from the supporting device(s) **120** may include the unique identifier to identify to which user input the response data corresponds.

[0174] In at least some embodiments, the user device **110** may include one or more skills **790** that may process similarly to the system-implemented skill(s) **690**. The skill(s) **790** installed on (or in communication with) the user device **110** may include, without limitation, a smart home skill and/or a device control skill configured to control a second device(s), a music skill configured to output music, a navigation skill configured to output directions, a shopping skill configured to conduct an electronic purchase, and/or the like.

[0175] FIG. **8** is a conceptual diagram of components of the smart home supporting device(s) **130** in addition to those described above. The smart home supporting device(s) **130** may include a voice user interface (VUI) component **805**, a graphical user interface (GUI) component **810**, a control layer **815**, an automations component **820**, a live model interface **825**, a streamed data processing component **830**, an offline model building component **835**, and one or more data storages **840**.

[0176] The VUI component **805** may include a smart home skill configured to process with respect to NLU output data corresponding to a home automation domain. The VUI component **805** may further include a component configured to handle ambiguous target resolution for home automation domain comments. This component may generate one or more questions for resolving an ambiguous entity reference in a spoken user input. For example, for the spoken user input “turn on couch”, this component may generate the question “did you mean the sofa light?” The VUI component **805** may include a recovery arbitration and friction feedback learning engine configured to arbitrate between misrouted smart home domain intents and, for example, music domain intents. The VUI component **805** may include a skill configured to handle group management (e.g., create/update membership/delete) and device rename intents. This skill may also attempt to recover user friction device turn on/off intents.

[0177] The GUI component **810** may include one or more front end services configured to interact with one or more APIs of an application implemented on a smart phone, tablet, or the like. The GUI component **810** may include a notification service configured to, for example, generate smart phone push notifications if a lock is left unlocked for longer than a threshold amount of time. The

GUI component **810** may include a recommended routines engine service configured to use a template mechanism to build routine recommendations to be displayed (e.g., in an application installed on a smart phone, tablet, or the like). The GUI component **810** may include a favorites component configured to provide device lists to be show in a “favorites” section of an application installed on a smart phone, tablet, or the like. The favorites component may also be configured to handle device starter set generation (e.g., call a ML model to determine which devices to show if no custom favorites are defined yet).

[0178] The control layer **815** may include an Internet of Things service configured to handle control and state retrieval of smart home devices. The control layer **815** may include a device state store service, which may be a cache of last known state of smart home devices. The control layer **815** may include a user home registry service configured to manage a database of all smart home devices and metadata, such as names, aliases, control routes, appliance types, etc. The control layer **815** may include a device metadata service configured to manage device metadata, such as language and location settings. The control layer **815** may include one or more settings tables, such as a permission and access control table and a favorites table. The control layer may include a device access API usable to provided record of current home configuration. The control layer may include a universal settings service configured to control various settings, such as change language, opt-in settings, etc.

[0179] The automations component **820** may include a triggers and sequences component configured to handle routines and other time- or event-triggered actions, such as sleep timers. The automations component **820** may include an arbitration component configured to perform automatic actions (e.g., turning off lights when a user is asleep). The automations component **820** may include a lighting timer service configured to turn lights on and off to present someone is home, as well as being configured to dim lights.

[0180] The live model interface **825** may include a targeting/entity resolution component that implements one or more targeting language models that rank entries as targets for home automation voice comments. The live model interface **825** may include a smart home machine learning model service configured to predict anomalies to send as VUI recommendations and/or mobile push recommendations. The live model interface may include a home state prediction component configured to implement an embedding-based model to predict expected home state. The live model interface **825** may include one or more models configured to filter recommendations based on past recommendation responses. The live model interface **825** may include a model endpoint to rank devices to be displayed as favorites for users who have not customized their favorite devices yet.

[0181] The streamed data processing component **830** may include a data ingestion service configured to ingest various data from various sources, store the data, and output said data for streamed and offline model processing. The streamed data processing component **830** may include an Internet of Things data warehouse that generates streams of data and performed periodic (e.g., weekly) dumps of data ingested by the data ingestion service. The streamed data processing component **830** may include a signal compute service configured to stream and join different data streams. The streamed data processing component **830** may include a raw activity service configured to compute user activity, such as “all users sleeping” or “someone active in the house. The output of the raw activity service may be used for automatic actions, such as turning off lights when everyone is sleeping. The streamed data processing component **830** may include a smart home automatized activity modeling service configured to compute activity transitions and send change events to subscribers. The streamed data processing component **830** may include a feedback computation service configured to perform feedback computation using sessionized data aggregated across time windows from multiple sources.

[0182] The offline model building component **835** may include a secure compute component configured to limit data egress and allow experimentation and computation on sensitive data. The

offline model building component **835** may include a scheduler service configured to schedule and run metrics or model update jobs. The offline model building component **835** may include a recommendations metrics job component configured to perform periodic (e.g., daily) recommendations metrics computations and response metrics. The offline model building component **835** may include periodic (e.g., daily) model update components configured to run streamed personalized feature updates. The offline model building component **835** may include a component configured to compute target inference priors (e.g., which devices are typically controlled at which time); a component configured to compute recommendation candidate lists, priors, and device embeddings; a component configured to compute embeddings and device features for favorites and personalized recommended routines; and/or a component configured to compute priors for typical activity by local time of day to be used in activity prediction.

[0183] The data storage(s) **840** may include a storage of service data that is consumed by machine learning pipelines to compute metrics, train models, and update per-user features. The data storage(s) **840** may include a segmented offline storage including hidden state data on a per-user level for periodic (e.g., daily) model updaters. The data storage(s) **840** may include one or more database tables for storing per-user and per-device features computed by offline job, to be used in live model inference. The data storage(s) **840** may include a self-service dataset generator that filters, transforms, and joins stream of data into batch-processing-friendly datasets.

[0184] As shown in FIGS. **1** and **8**, the smart home supporting device(s) **130** may also include the device control application **140**, the GSS **145** and the device group storage **165**. As described herein, the device control supporting device(s) **135** may be implemented as a skill supporting device(s) **625**. When a user input relating to a device group is received via the device control application **140**, the smart home supporting device(s) **130** may communicate with one or more skill supporting device(s) **625** to cause operation of a device(s) **110**, **112** included in the device group. For example, as described in relation to FIGS. **4** and **5**, the smart home supporting device(s) **130** may communicate device group changes to or may receive device group changes from the skill supporting device(s) **625**. As another example, the device control application **140** may receive a user input to operate the devices **110**, **112** in a device group, where at least a first device **110** of the device group is controlled by the skill **625a**. In such cases, the smart home supporting device(s) **130** may send data indicative of the user input to the skill **625a** along with a group identifier used by the skill **625a** to identify the device group. In response to receiving the data, the skill **625a** may retrieve device group data associated with the group identifier, and cause operation of the first device included in the device group.

[0185] FIG. **9** is a block diagram conceptually illustrating a device **110** that may be used with the supporting device(s) **120** and/or the smart home supporting device(s) **130**. FIG. **10** is a block diagram conceptually illustrating example components of a device, such as the supporting device(s) **120**, the smart home supporting device(s) **130**, the device control supporting device(s) **135**, and/or the skill supporting device(s) **625**. The supporting device(s) **120**, **130**, **135**, **625** may include one or more servers. A “server” as used herein may refer to a traditional server as understood in a server/client computing structure but may also refer to a number of different computing components that may assist with the operations discussed herein. For example, a server may include one or more physical computing components (such as a rack server) that are connected to other devices/components either physically and/or over a network and is capable of performing computing operations. A server may also include one or more virtual machines that emulates a computer system and is run on one or across multiple devices. A server may also include other combinations of hardware, software, firmware, or the like to perform operations discussed herein. The supporting device(s) **120**, the smart home supporting device(s) **130**, the device control supporting device(s) **135**, and/or the skill supporting device(s) **625** may be configured to operate using one or more of a client-server model, a computer bureau model, grid computing techniques, fog computing techniques, mainframe techniques, utility computing techniques, a peer-to-peer

model, sandbox techniques, or other computing techniques.

[0186] Multiple supporting devices **120** may be included in the system **100** of the present disclosure, such as one or more supporting devices **120** for performing ASR processing, one or more supporting devices **120** for performing NLU processing, and one or more skill components, etc. In operation, each of these systems may include computer-readable and computer-executable instructions that reside on the respective supporting device(s) **120**, as will be discussed further below.

[0187] Each of these devices (**110/120/130/135/625**) may include one or more controllers/processors (**904/1004**), which may each include a central processing unit (CPU) for processing data and computer-readable instructions, and a memory (**906/1006**) for storing data and instructions of the respective device. The memories (**906/1006**) may individually include volatile random access memory (RAM), non-volatile read only memory (ROM), non-volatile magnetoresistive memory (MRAM), and/or other types of memory. Each device (**110/120/130/135/625**) may also include a data storage component (**908/1008**) for storing data and controller/processor-executable instructions. Each data storage component (**908/1008**) may individually include one or more non-volatile storage types such as magnetic storage, optical storage, solid-state storage, etc. Each device (**110/120/130/135/625**) may also be connected to removable or external non-volatile memory and/or storage (such as a removable memory card, memory key drive, networked storage, etc.) through respective input/output device interfaces (**902/1002**).

[0188] Computer instructions for operating each device (**110/120/130/135/625**) and its various components may be executed by the respective device's controller(s)/processor(s) (**904/1004**), using the memory (**906/1006**) as temporary “working” storage at runtime. A device's computer instructions may be stored in a non-transitory manner in non-volatile memory (**906/1006**), storage (**908/1008**), or an external device(s). Alternatively, some or all of the executable instructions may be embedded in hardware or firmware on the respective device in addition to or instead of software.

[0189] Each device (**110/120/130/135/625**) includes input/output device interfaces (**902/1002**). A variety of components may be connected through the input/output device interfaces (**902/1002**), as will be discussed further below. Additionally, each device (**110/120/130**) may include an address/data bus (**924/1024**) for conveying data among components of the respective device. Each component within a device (**110/120/130/135/625**) may also be directly connected to other components in addition to (or instead of) being connected to other components across the bus (**924/1024**).

[0190] Referring to FIG. 9, the device **110** may include input/output device interfaces **902** that connect to a variety of components such as an audio output component such as a speaker **912**, a wired headset or a wireless headset (not illustrated), or other component capable of outputting audio. The device **110** may also include an audio capture component. The audio capture component may be, for example, a microphone **920** or array of microphones, a wired headset or a wireless headset (not illustrated), etc. If an array of microphones is included, approximate distance to a sound's point of origin may be determined by acoustic localization based on time and amplitude differences between sounds captured by different microphones of the array. The device **110** may additionally include a display **916** for displaying content. The device **110** may further include a camera **918**.

[0191] Via antenna(s) **914**, the input/output device interfaces **902** may connect to one or more networks **199** via a wireless local area network (WLAN) (such as Wi-Fi) radio, Bluetooth, and/or wireless network radio, such as a radio capable of communication with a wireless communication network such as a Long Term Evolution (LTE) network, WiMAX network, 3G network, 4G network, 5G network, etc. A wired connection such as Ethernet may also be supported. Through the network(s) **199**, the system may be distributed across a networked environment. The I/O device

interface (902/1002) may also include communication components that allow data to be exchanged between devices such as different physical servers in a collection of servers or other components. [0192] The components of the device 110, the supporting device(s) 120, the smart home supporting device(s) 130, and/or the skill supporting device(s) 625 may include their own dedicated processors, memory, and/or storage. Alternatively, one or more of the components of the device 110, the supporting device(s) 120, the smart home supporting device(s) 130, and/or a skill supporting device(s) 625 may utilize the I/O interfaces (902/1002), processor(s) (904/1004), memory (906/1006), and/or storage (908/1008) of the device(s) 110, supporting device(s) 120, the smart home supporting device(s) 130, or the skill supporting device(s) 625, respectively.

[0193] As noted above, multiple devices may be employed in a single system. In such a multi-device system, each of the devices may include different components for performing different aspects of the system's processing. The multiple devices may include overlapping components. The components of the device 110, the supporting device(s) 120, the smart home supporting device(s) 130, and the skill supporting device(s) 625, as described herein, are illustrative, and may be located as a stand-alone device or may be included, in whole or in part, as a component of a larger device or system.

[0194] As illustrated in FIG. 11, multiple devices may contain components of the system 100 and the devices may be connected over a network(s) 199. The network(s) 199 may include a local or private network or may include a wide network such as the Internet. Devices may be connected to the network(s) 199 through either wired or wireless connections. For example, a speech controllable device 110a, a smart phone 110b, a smart watch 110c, a tablet computer 110d, a speech-controlled device 110f with a display, a television 110g, a washer/dryer 110h, a refrigerator 110i, a microwave 110j, an outlet 112b, a light bulb 112a, a light switch 112c, and a camera(s) 112d may be connected to the network(s) 199 through a wireless service provider, over a Wi-Fi or cellular network connection, or the like. Other devices that may be connected to the network(s) 199 through either wired or wireless connections include a lock, a fan, a WIFI router, a thermostat, temperature sensor, a garage door, an alarm system, a motion sensor, a contact sensor, a smoke alarm, etc. Other devices are included as network-connected support devices, such as the supporting device(s) 120, the smart home supporting device(s) 130, and/or others. The support devices may connect to the network(s) 199 through a wired connection or wireless connection. Networked devices may capture audio using one-or-more built-in or connected microphones or other audio capture devices, with processing performed by ASR components, NLU components, or other components of the same device or another device connected via the network(s) 199.

[0195] The concepts disclosed herein may be applied within a number of different devices and computer systems, including, for example, general-purpose computing systems, speech processing systems, and distributed computing environments.

[0196] The above aspects of the present disclosure are meant to be illustrative. They were chosen to explain the principles and application of the disclosure and are not intended to be exhaustive or to limit the disclosure. Many modifications and variations of the disclosed aspects may be apparent to those of skill in the art. Persons having ordinary skill component in the field of computers and speech processing should recognize that components and process steps described herein may be interchangeable with other components or steps, or combinations of components or steps, and still achieve the benefits and advantages of the present disclosure. Moreover, it should be apparent to one skilled in the art, that the disclosure may be practiced without some or all of the specific details and steps disclosed herein.

[0197] Aspects of the disclosed system may be implemented as a computer method or as an article of manufacture such as a memory device or non-transitory computer readable storage medium. The computer readable storage medium may be readable by a computer and may comprise instructions for causing a computer or other device to perform processes described in the present disclosure. The computer readable storage medium may be implemented by a volatile computer memory, non-

volatile computer memory, hard drive, solid-state memory, flash drive, removable disk, and/or other media. In addition, components of system may be implemented as in firmware or hardware, such as an acoustic front end (AFE), which comprises, among other things, analog and/or digital filters (e.g., filters configured as firmware to a digital signal processor (DSP)).

[0198] Conditional language used herein, such as, among others, “can,” “could,” “might,” “may,” “e.g.,” and the like, unless specifically stated otherwise, or otherwise understood within the context as used, is generally intended to convey that certain embodiments include, while other embodiments do not include, certain features, elements and/or steps. Thus, such conditional language is not generally intended to imply that features, elements, and/or steps are in any way required for one or more embodiments or that one or more embodiments necessarily include logic for deciding, with or without other input or prompting, whether these features, elements, and/or steps are included or are to be performed in any particular embodiment. The terms “comprising,” “including,” “having,” and the like are synonymous and are used inclusively, in an open-ended fashion, and do not exclude additional elements, features, acts, operations, and so forth. Also, the term “or” is used in its inclusive sense (and not in its exclusive sense) so that when used, for example, to connect a list of elements, the term “or” means one, some, or all of the elements in the list.

[0199] Disjunctive language such as the phrase “at least one of X, Y, Z,” unless specifically stated otherwise, is understood with the context as used in general to present that an item, term, etc., may be either X, Y, or Z, or any combination thereof (e.g., X, Y, and/or Z). Thus, such disjunctive language is not generally intended to, and should not, imply that certain embodiments require at least one of X, at least one of Y, or at least one of Z to each be present.

[0200] As used in this disclosure, the term “a” or “one” may include one or more items unless specifically stated otherwise. Further, the phrase “based on” is intended to mean “based at least in part on” unless specifically stated otherwise.

Claims

1. A computer-implemented method comprising: receiving first input data associated with a user profile and representing authorization to synchronize device group information across different device control applications; receiving first device group data associated with the user profile and with a first device control application, the first device group data representing a first device; receiving second device group data associated with the user profile and with a second device control application, the second device group data representing a second device; in response to receiving the first input data, processing the first device group data and the second device group data to determine third device group data representing the first device and the second device; and enabling at least one of the first device control application and the second device control application to control a device group using the third device group data.
2. The computer-implemented method of claim 1, wherein processing the first device group data and the second device group data comprises processing the first input data, the first device group data, and the second device group data using a machine learning model to determine the third device group data.
3. The computer-implemented method of claim 1, wherein: the first device group data corresponds to a first device group name; and the second device group data corresponds to a second device group name similar to the first device group name.
4. The computer-implemented method of claim 1, wherein: the first device group data represents a third device; the second device group data represents the third device; and the third device group data represents the third device.
5. The computer-implemented method of claim 1, wherein: the first device group data corresponds to a first location; and the second device group data corresponds to the first location.

- 6.** The computer-implemented method of claim 1, further comprising: sending, to the first device control application, the third device group data to use instead of the first device group data.
- 7.** The computer-implemented method of claim 6, further comprising: sending, to the second device control application, the third device group data to use instead of the second device group data.
- 8.** The computer-implemented method of claim 6, wherein the first input data corresponds to a user permission associated with the user profile, and wherein the third device group data is sent to the first device control application in absence of a user expressly indicating the first device group data is to be updated.
- 9.** The computer-implemented method of claim 1, further comprising: determining the first device has become associated with the first device group data, wherein determination of the third device group data is based at least in part on the first device becoming associated with the first device group data.
- 10.** The computer-implemented method of claim 1, further comprising: determining that a third device has become disassociated with the first device group data, wherein determination of the third device group data is based at least in part on the third device becoming disassociated with the first device group data and wherein the third device group data does not represent the third device.
- 11.** A system comprising: at least one processor; and at least one memory including instructions that, when executed by the at least one processor, cause the system to: receive first input data associated with a user profile and representing authorization to synchronize device group information across different device control applications; receive first device group data associated with the user profile and with a first device control application, the first device group data representing a first device; receive second device group data associated with the user profile and with a second device control application, the second device group data representing a second device; in response to receipt of the first input data, process the first device group data and the second device group data to determine third device group data representing the first device and the second device; and enable at least one of the first device control application and the second device control application to control a device group using the third device group data.
- 12.** The system of claim 11, wherein the instructions that cause the system to process the first device group data and the second device group data comprise instructions that, when executed by the at least one processor, cause the system to process the first input data, the first device group data, and the second device group data using a machine learning model to determine the third device group data.
- 13.** The system of claim 11, wherein: the first device group data corresponds to a first device group name; and the second device group data corresponds to a second device group name similar to the first device group name.
- 14.** The system of claim 11, wherein: the first device group data represents a third device; the second device group data represents the third device; and the third device group data represents the third device.
- 15.** The system of claim 11, wherein: the first device group data corresponds to a first location; and the second device group data corresponds to the first location.
- 16.** The system of claim 11, wherein the at least one memory includes further instructions that, when executed by the at least one processor, further cause the system to: send, to the first device control application, the third device group data to use instead of the first device group data.
- 17.** The system of claim 16, wherein the at least one memory includes further instructions that, when executed by the at least one processor, further cause the system to: send, to the second device control application, the third device group data to use instead of the second device group data.
- 18.** The system of claim 16, wherein the first input data corresponds to a user permission associated with the user profile, and wherein the third device group data is sent to the first device control application in absence of a user expressly indicating the first device group data is to be updated.
- 19.** The system of claim 11, wherein the at least one memory includes further instructions that,

when executed by the at least one processor, further cause the system to: determine the first device has become associated with the first device group data, wherein determination of the third device group data is based at least in part on the first device becoming associated with the first device group data.

20. The system of claim 11, wherein the at least one memory includes further instructions that, when executed by the at least one processor, further cause the system to: determine that a third device has become disassociated with the first device group data, wherein determination of the third device group data is based at least in part on the third device becoming disassociated with the first device group data and wherein the third device group data does not represent the third device.
